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What Can Satellite Imagery and Machine Learning Measure?

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ABSTRACT

Satellite imagery and machine learning (SIML) are increasingly being combined to remotely measure social and environmental outcomes, yet use of this technology has been limited by insufficient understanding of its strengths and weaknesses. Here, we undertake the most extensive effort yet to characterize the potential and limits of using a SIML technology to measure ground conditions. We conduct 115 standardized large-scale experiments using a composite high-resolution optical image of Earth and a generalizable SIML technology to evaluate what can be accurately measured and where this technology struggles. We find that SIML alone predicts roughly half the variation in ground measurements on average, and that variables describing human society (e.g. female literacy, $R^2=0.55$) are generally as easily measured as natural variables (e.g. bird diversity, $R^2=0.55$). Patterns of performance across measured variable type, space, income and population density indicate that SIML can likely support many new applications and decision-making use cases, although within quantifiable limits.

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Website homepage for new satellite imagery
embeddings is available at:
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Rapidly growing satellite imagery archives are transforming the way we map, monitor, and understand our planet. The promise of this new information is that it can be harnessed to create comprehensive and detailed records of many aspects of human well-being and environmental conditions that might ultimately transform decision-making and societies (1). A major challenge to utilizing these data is the extraordinary volume of incoming information, now amounting to hundreds of terabytes collected daily. Automated algorithms are well-suited to distill this flood of imagery into structured datasets that are usually both more interpretable by humans and immediately applicable to decision-making. For example, individual studies have demonstrated that satellite imagery coupled with machine learning (SIML) can skillfully construct maps of forest cover (2), income (3), flooding (4), solar panels (5), croplands (6), and housing prices (7), along with numerous other individual outcomes; and recent developments in foundation models for earth observation have the potential to increase the pace of this satellite-based mapping (e.g., (7–10)). Yet it remains unclear for which variables and under what conditions imagery is a valuable source of information about ground conditions. If SIML is effective in these individual published cases, what else can it be used for? And perhaps more importantly, what shouldn't it be used for?

A core obstacle to the broader adoption and utilization of SIML technologies are ambiguities around its abilities and limitations. Potential users that depend on structured geospatial information may not know if SIML is likely to be able to measure the quantities they require nor how reliable such information might be. Gleaning this information from published studies (if the quantities of interest are not already the subject of one) is difficult, since published studies generally focus on highly successful applications of custom systems that extract a single variable in a particular context. Though the potential and limits of SIML have been tested in specific geographies (7, 11–14) a systematic examination of the opportunities and limitations of imagery-based prediction is needed. Demand for such understanding is underscored both by large investments (by governments, researchers, the private sector, and community organizations) in satellite-based infrastructure as well as by persistent skepticism and resistance to adopt SIML technologies in some sectors where it might potentially generate widespread societal gains (15). Without a clearer and more general understanding of where and when SIML works, we may risk both over- and/or under-investing in SIML systems.

Here, we undertake what is to our knowledge the most extensive effort yet to characterize the potential and limits of using a SIML technology to measure social and environmental outcomes. Specifically, we quantitatively assess what types of information can be recovered from high-resolution optical imagery across national and global scales by conducting 115 parallel experiments that use a single composite image of Earth (16) and the same

standardized machine learning procedure (7). To do this, we assemble a new benchmark dataset (“*SIMLbench*”) containing cross-sectional ground-condition information for 115 diverse variables at national- to global-scale (hereafter, “ground truth”, with the qualification that some of these measurements are themselves modeled) that are all linked to a common dataset of features from tri-band (i.e., red, green, and blue) imagery. SIMLbench contains datasets describing human health, demographics, built infrastructure, occupation, agriculture, education, income, natural systems, household assets, and agricultural assets. Each experiment attempts to measure one variable from SIMLbench using only the information contained in the imagery dataset, resulting in a new SIML-derived map. We then evaluate these maps against the ground truth measurements in a held-out test set to identify the types of variables and locations in which such measurements can be made with high fidelity and in which errors are larger. Importantly, SIMLbench is constructed from a diversity of data sources with differing extents, resolutions, and data qualities. Such diversity reflects the types of training data used by many remote sensing practitioners, but can also complicate comparisons of model performance across variables and regions (see Materials and Methods Section A.1 and Supplementary Materials Section D).

In addition to collecting a large dataset of different ground conditions, two other core obstacles to understanding what is measurable by SIML have been (i) the computational burden of running many large parallel experiments and (ii) the diversity – and incomparability – of ML methodologies employed in the literature. We overcome both of these challenges by employing a single standardized experimental protocol that utilizes a generalizable and accessible ML method designed to solve multi-task problems efficiently (Materials and Methods Section A.3). Each sample of ground conditions (Y) is linked to a $0.01^\circ \times 0.01^\circ$ ($\sim 1\text{km}^2$) global grid that is aligned with a single cross-section of tri-band optical imagery (I). We then use the Multi-task Observation using Satellite Imagery & Kitchen Sinks (MOSAICS) method (7) to transform the imagery into 4,000 task-agnostic features ($m(I) \rightarrow X$) which are in turn used to learn and predict the outcome at all locations ($f(X) \rightarrow \hat{Y}$). MOSAICS is well suited for this experimental context because the computationally costly step of transforming the imagery into features can be reused without alteration in all 115 parallel experiments. MOSAICS features represent an over-complete basis for embedding imagery, which prior work demonstrated was both highly descriptive and generalized across tasks (7). Each experiment using MOSAICS is orders of magnitude faster and less resource intensive than training and applying a deep-learning approach like a convolutional neural network, but its performance remains competitive with these deep learning approaches and related methods, as shown in prior work (7, 8) and the analysis below. This similarity in performance suggests that our findings are likely to generalize across many SIML methods. We note that

other generalizable satellite imagery embeddings, such as AlphaEarth Foundations (8) or Galileo (9), could be used within this same experimental setup; we choose to use MOSAIKS as a representative feature embedding of satellite imagery due to its simplicity, public availability of features globally, and correspondence in performance with other approaches (see Discussion). In addition to a single ML methodology, we use a single satellite image source across all tasks (2019 Planet Labs cloud-free composite). We use high-resolution ($\approx 5\text{m}$) optical imagery due to its widespread use and its demonstrated ability to capture textural information relevant to both socioeconomic and environmental variables, including those like income that do not have a known spectral signature (1, 11, 17–20). While more complex or variable-specific model training and imagery selection could theoretically improve predictive performance, our results present a lower bound of how well variables can be predicted with satellite imagery.

We also note that these imagery-based predictions represent an expansion of the social, economic, and environmental conditions that have previously been mapped from imagery. Of the 115 variables we investigate, 56 have been previously predicted with satellite imagery alone, while 48 have never been estimated at the national or global scale using satellite imagery.

This assessment provides actionable information to guide the development and use of SIML technology. In addition, it reveals potential vulnerabilities in the technology and identifies directions for improvement, which are critical features to uncover as imagery-based systems are increasingly used to provide information in data poor and/or inaccessible regions (1, 21, 22). To facilitate the science and practice of imagery-based mapping generally, we publicly distribute SIMLbench, the 115 maps produced in our experiments, as well as spatially differentiated measures of predictive performance. We also release the featurized image of Earth and the generalizable code that should allow any researcher to conduct a comparable experiment (and generate a new map) based on their own ground truth data.

Results

115 experiments generating maps from optical imagery

Using a single composite optical image of Earth, we use the same MOSAIKS procedure to independently train a model to predict each of 115 outcomes from a single set of image features (see Materials and Methods Sections A.3 and A.4). The output from each experiment is a large-scale map. This results in a sample of 115 maps describing social, economic, and environmental variables at 0.01° resolution that we analyze (Figure 1). Our focus is

evaluating SIML technology, however these maps also could be used to provide a rich assessment of ground conditions within a single location, ranging from drinking water access to irrigation infrastructure (Table SM1). Our objective here is to characterize the quality of this information.

Visual inspection These maps display intuitive spatial patterns. For example, maps of per-capita income show pockets of high income around cities, which are particularly visible in the Global South due to low income levels in surrounding rural areas. Within sub-Saharan Africa, the Middle East and western Asia, areas with high male employment in agriculture tend to also have high childhood stunting and low female literacy. Livestock ownership is noticeably high in the Sahel, where cattle, sheep, goat, and poultry provide a key source of income and protection against climate vulnerability (23). In northern India, where cycling serves as a central mode of urban transport, bike ownership is high (24). Irrigation’s ability to make deserts bloom is visible in the agriculturally rich Indus river valley and Nile delta.

Explanatory power of imagery Imagery cannot perfectly recover ground conditions, since many crucial types of information cannot be observed from satellites, a fact that will create an unknown variable-specific “ R^2 -ceiling” on each map’s accuracy. On average across the 115 experiments, we find that satellite imagery explains 45% of the spatial variation in held-out test set ground measurements (Table 1; data held out randomly to measure map accuracy in areas similar to the training data, as described in Materials and Methods Section A.4; see Figure 3 for spatial extrapolation performance measured by spatial cross-validation). For 46% of experiments, imagery explains over half of the spatial variation, and for 17% it explains over two-thirds of the variation. Predictive power is notably high for many variables that have never before been predicted at scale from imagery alone, such as vehicle ownership, agricultural involvement, and life expectancy, with R^2 values of 0.66, 0.62, and 0.72, respectively. Other variables are more difficult to discern from imagery: for 10% of the experiments, imagery explains less than 10% of the spatial variation, and for 23% of experiments it explains less than 25%.

Comparison of generalizable SIML to prior benchmarks The performance of these MOSAIKS-based predictions is similar to that of previous approaches developed for specific tasks, suggesting our insights are likely to generalize to other methods of imagery-based prediction. For the 53 experiments that have comparable benchmarks in prior work, our predictive performance correlates strongly with performance from existing studies ($r = 0.47, p < 0.001$), Figure SM1). Consistent with prior findings (7), MOSAIKS performs competitively

with prior approaches, exceeding the performance of comparable estimates from the literature in 33 out of the 53 cases. Of course, direct comparisons with the literature are complicated by the fact that many prior studies use different source data and employ different procedures for training and evaluation (Materials and Methods Section A.5). Despite this, competitive performance that correlates well with comparable benchmarks using a range of approaches indicates the usefulness of this experiment for informing how satellite-based models perform across a range of variables and contexts.

What is measurable with SIML?

SIML experiments had the highest performance for variable categories related to agriculture, household assets, and natural systems, with average R^2 s of 0.54-0.60 (Figure 2a). Variables related to demographics, education, and income exhibit intermediate performance, with average R^2 s of 0.45-0.51. Variables related to built infrastructure, agricultural assets, health, and occupation are, on average, the most difficult to detect with imagery, with average R^2 s of 0.23-0.41. Individual variables and their categories, performance, and other properties are listed in Table SM2.

Skill within and across outcome categories Across all 115 experiments, 32% of the variance in performance is explained by the ten variable categories shown in Figure 2a. The variance in predictive performance *within* categories is of similar magnitude to variance *across* categories ($\sigma_{within} = 0.19$ and $\sigma_{across} = 0.12$). This suggests that factors other than the variable category also influence how amenable a given variable is to imagery-based prediction. For example, while maize yields can be predicted with relatively high accuracy ($R^2 = 0.90$), the fraction of land used for tree cropping, another agricultural variable, is much more difficult to measure ($R^2 = 0.16$), possibly because tree crops may appear visually similar to natural forests. Similarly, while direct livelihood measures such as gross national income ($R^2 = 0.72$) and the United Nations Human Development Index ($R^2 = 0.78$) perform relatively well, variables that have a more complex relationship with economic well-being, such as the income-to-poverty ratio ($R^2 = 0.21$) or percent of households rejected for a loan ($R^2 = 0.04$), prove much more difficult to predict with optical imagery. Finally, variance in performance is particularly high within the built infrastructure category, where the fraction of land area covered by buildings can be estimated relatively well ($R^2 = 0.78$), but the length of petroleum or crude oil pipelines is not detectable with imagery ($R^2 < 0.01$). In this case, visibility of the infrastructure is likely critical: buildings are directly visible in imagery, while pipelines are generally buried underground and therefore not possible to directly identify with imagery (Figure SM8).

Visibility The importance of direct visibility in determining SIML performance holds across the full set of experiments: variables that are directly represented in the imagery – such as forest cover, cropped area, or road length – are on average easier to measure than variables that either must be inferred from objects visible in the image, such as income, or are not visible at all, such as below-ground pipelines (Figure 2b; difference in mean $R^2 = 0.18$; $p < 0.001$). This pattern partially explains, for example, why the agriculture category is high-performing, as 94% of variables in this category are directly visible. However, we also achieve high predictive performance on many variables that are not directly visible, such as schooling, measures of income and assets, and life expectancy, suggesting that a variable need not be visible in the imagery for it to be predictable using imagery. We determined whether variables are considered directly visible or not prior to the analysis (Table SM2).

Natural vs. anthropocentric variables We find no systematic difference between performance for variables directly related to human activity, such as educational attainment and mobile phone ownership, and those not directly related to human activity, such as tree species richness and elevation (Figure 2c; difference in mean $R^2 = 0.1$, $p = 0.23$). This suggests that the historical emphasis in remote sensing on natural systems is not indicative of fundamental limitations of applying these tools to human systems. We also find that variables with a clear seasonal cycle are higher-performing than those without, despite our model taking in only a single snapshot of Earth (Figure 2d; mean difference in $R^2 = 0.19$, $p < 0.05$), and that spatial resolution of the ground truth data has no systematic relation to performance (Figure 2e; all differences in R^2 across resolution categories yield $p > 0.1$).

SIML accuracy in regions far from training data

In each experiment, we evaluate how SIML performance changes as the distance between training data and predictions increases. Prior work indicates that SIML map accuracy will tend to decline as predictions are made further from training observations (7, 22, 25). Systematic assessment of SIML to generalize across space is therefore essential to supporting operational use of SIML technologies. Specifically, in each experiment we split the training and validation data into spatially disjoint sets following a checkerboard pattern, training on the “black squares” and predicting on the “white squares”, as shown in Figure 3k (following ref. (7)). As the checkerboard size increases from 2.5° ($\sim 275\text{km}$) to 16° ($\sim 1650\text{km}$), a greater degree of spatial extrapolation is required from the training to the validation locations (Materials and Methods Section A.5).

As expected, we find that map accuracy generally decreases with increased distance between training and validation data (Figure 3a-j). Across all variables, median performance

drops by 50% (from median $R^2=0.44$ to 0.22) as separation between training and validation data increases from no prescribed separation to a checkerboard of width 16° ($\sim 1650\text{km}$). However, variables that are relatively well-predicted by satellite imagery overall tend to lose a smaller fraction of performance when spatially extrapolating than those that are poorly predicted. For example, housing density performance falls by just 3.5% (from $R^2=0.85$ to 0.82) between no spatial extrapolation and a checkerboard of width 16° , while neonatal mortality, a more difficult variable to predict from imagery, falls by 81% (from $R^2=0.07$ to 0.01) in this same exercise.

Direct visibility and spatial generalizability We find that experiments with variables that are directly visible in the image (Figure 2b) usually enable a more robust model for spatial extrapolation than variables that rely on proxy relationships between the visible objects and the variable of interest, a relationship which may change across space. Median performance predicting variables that are not directly visible falls by 72% (from $R^2=0.39$ to 0.11) in the 16° checkerboard test, while directly visible variables fall only 28% (from $R^2=0.64$ to 0.46) (Figure SM4). In this test, the value of direct visibility for spatial extrapolation is apparent within variable categories as well as across them. For example, within built infrastructure, estimates of directly visible road length lose little performance extrapolating over space (R^2 declines by 10%), whereas estimates of housing unit vacancy lose roughly half their predictive performance (Figure SM9). In general, performance decays most rapidly with distance for variables in the agricultural assets and income categories, but least rapidly for variables in the agriculture and demographics categories (the agricultural assets category assesses the value of assets like livestock, while the agriculture category indicates the presence of agricultural activity or the productivity of crops; see Table 1) .

Context-dependent SIML accuracy

Are certain geographic contexts more or less amenable to mapping with SIML? For example, poor training data quality or complex visual signals could make imagery-based prediction more difficult in developing country contexts (1) or in rural areas (26). To answer this question, we aggregate information across all experiments to evaluate SIML performance *in general* at each location. In each location, we aim to understand how one might expect SIML to perform for a large collection of diverse variables, or for an unknown variable. We do this by summarizing performance across experiments at each location while also accounting for the uneven availability of ground truth data as well as different units of measurement and spatial variances. We construct a new metric for evaluating model errors across multiple experiments called “mean absolute normalized error” (MANE). Analogous to a z -score, the

value of MANE in a location indicates the average normalized (i.e., $\mu = 0, \sigma = 1$) prediction error across all experiments with available ground truth data in that location (Materials and Methods Section A.5). The purpose of MANE is to identify *locations* where image-based measurement is, on average across all the variables we study, higher- versus lower-performing in general. For each location, MANE quantifies an average measure *across all experiments* of standardized performance *within each experiment*. A map of MANE for all variables is shown in Figure 4a and maps of MANE for each variable category are shown in Figures SM12-SM14.

Accuracy across space We find strong evidence that SIML accuracy in general differs across continents (Figure 4a, inset panel, $p < 0.01$ for all pairwise comparisons). Prediction errors are lowest in Australia and Africa, where large, unpopulated deserts exhibit low MANE. Errors are in general highest in North America and Asia, where urban areas and high elevation regions are prevalent and exhibit higher error. These results suggest that poor image quality or limited training data in the world’s low-income regions are not dominant forces in predictability across large spatial scales. Instead, geographic features of the landscape appear to be most important when assessing imagery-based model performance across large global regions. Specifically, locations with extreme values, such as mountainous areas with low population density and high elevation or metropolitan areas with high population density and road length, tend to have higher MANE. For example, global average MANE is 0.67 but MANE over the high-elevation Tibetan Plateau averages to 1.28 and across the 10 largest global cities averages to 1.08. This finding is consistent with the general tendency of machine learning models to exhibit mean-reverting measurement error, where relatively high values are underestimated and relatively low values are overestimated (27, 28). We show additional visual evidence of such mean-reverting measurement error (i.e., over-prediction of low values and under-prediction of high values) in Figure SM2, where predictions are plotted against ground truth for all 115 experiments.

Accuracy by income and population density We find that for most variable categories, MANE varies systematically with income and/or population density (Figure 4b). Specifically, MANE tends to be either constant or to decline with income, implying that higher-income locations tend to have lower imagery-based model error (Figure 4b). This is particularly true for agricultural, educational, health, and occupation variable categories. For example, average MANE for health variables falls by 40% between the lowest and highest income global grid cells. In contrast, errors in demographic and household asset variables exhibit little relationship with income, while in some cases, model error tends to *increase*

with income (e.g., agricultural assets). Similarly, MANE tends to increase with population density, suggesting imagery-based prediction error is higher in urban areas and large cities for many variables, as evident in Figure 4a. For example, MANE for infrastructure-related variables more than quadruples from 0.5 to over 2.0 between sparsely populated and densely populated grid cells. These patterns may be due to higher extreme values and a lower proportion of visual signal (such as seeing only the top of an apartment building but not each individual home) in densely populated areas. The opposite patterns, however, are found for variables related to health, demographics, and household assets, where prediction error declines with population density.

Public distribution of imagery-based maps and SIMLbench data

To our knowledge, our data represents the largest set of variables mapped in a single analysis using high-resolution optical imagery. We are the first to compute satellite-derived estimates of variables ranging from occupational outcomes to life expectancy to livestock ownership at the national to global scale. We publicly release predictions for all 115 variables across global land areas at $0.01^\circ \times 0.01^\circ$ resolution (Figure 1), accompanied by data quality layers that provide location- and variable-specific measures of estimated model error. Data quality scores are derived directly from the experiment-specific spatial generalization test results shown in Figure 3, such that locations near to ground truth receive higher quality scores than those farther away (Materials and Methods Section A.6). Figure SM3 illustrates how these quality scores are calculated using household access to flush toilets as an example, while quality scores across all variables for two example images are shown in Table SM1.

Of course, we expect that not all of the 115 maps we release will be of practical value. For example, this analysis provides users interested in variables that are already measured at global scale, such as terrain ruggedness or forest cover, information on how well they can be measured using optical satellite imagery, but practitioners should use data from the original source. It is also unlikely that maps with relatively poor performance – such as the 19 variables where our predictions explain less than a fifth of the ground truth variation – will be useful in many applications.

However, many of our maps have relatively high performance and represent the first satellite-based estimates of a variable. For example, 36 maps have an R^2 greater than 0.45 for variables that have never been remotely sensed at national or global scale. These variables include: the fraction of men and of women employed in agriculture ($R^2 = 0.45$ and 0.47); mobile and landline phone ownership ($R^2 = 0.51$ and 0.52); water access ($R^2 = 0.57$); life expectancy ($R^2 = 0.72$); food worrying ($R^2 = 0.49$); expected and obtained years of

education ($R^2 = 0.65$, and 0.72); female and male literacy ($R^2 = 0.55$ and 0.40); refrigerator, bike, and livestock ownership ($R^2 = 0.72$, 0.68 , and 0.61); household rent ($R^2 = 0.53$); and electricity access ($R^2 = 0.67$). The relatively high performance of the MOSAIKS approach across many variables indicates that it generalizes to a substantially broader set, by an order of magnitude, of variables than previously shown (7).

We also release all SIMLbench ground truth data (that are not protected by data-use agreements) as a benchmark data set for use by the SIML research community. We use MOSAIKS in this analysis because it is computationally efficient and generalizable, aspects that are important for our standardized comparison across experiments, but we expect that other approaches applied to SIMLbench will eventually improve upon the performance that we report.

Satellite-based maps and data quality layers, alongside all input data including satellite features and SIMLbench ground truth data, as well as our code, are hosted at <http://www.mosaiks.org>. (Note: these will be made publicly available upon publication.)

Discussion

Due to growing availability of satellite data, imagery-based mapping efforts are increasingly being developed and deployed in research and practice to measure and monitor our planet’s people, places, and economies. However, this technology is costly to construct and tailor to any given context, and investments are being made without a systematic understanding of which variables and in which locations predictions are likely to be reliable. This paper provides an extensive analysis of the types of variables and locations in which satellite remote sensing approaches based on high-resolution optical imagery are likely to be accurate. We find that variables directly represented in the imagery tend to have higher performance, especially when spatially extrapolating; human-made variables are generally as easily measured as natural variables; and errors in predictions differ systematically across space and tend to be associated with income and population.

Our results may help users of SIML-based data products interpret their quality and more responsibly use them for scientific inference. For example, our findings that SIML-based predictions exhibit substantial performance differences across space and by income and population raise important considerations for downstream use of these data in statistical models (28). Moreover, our identification of regions with high prediction error may help avoid the misallocation of social policies or interventions when imagery-based predictions are used to inform deployment of public resources (26).

Our analysis has several important limitations. First, our pipeline enables production of

many satellite-based maps by imposing a degree of standardization and automation (Materials and Methods Section A.4, Supplementary Materials Section E). It is likely that more customized treatment of individual variables, including custom satellite features, additional sensors, or alternative modeling strategies not considered here, could improve performance. To support future advances, our results can be used to guide such investments in more costly, customized models, and our data release can serve as a benchmark dataset for evaluation of other approaches.

Second, our analysis uses only a single imagery source and embedding. To explore potential differences in patterns of performance using alternative satellites, image spectra, and machine learning techniques, we repeat eight of our experiments using the recently released AlphaEarth Foundations (AEF) features in place of MOSAIKS features (see Methods A.5). AEF features were created using optical, thermal and C-band synthetic aperture radar information from Sentinel 1 and 2 and Landsat 8 and 9 satellite imagery and have been shown in prior work to have state-of-the-art performance compared with a range of other satellite image embeddings (8). We find relatively similar patterns of performance between models using these two types of satellite features: maize yield ($R_{AEF}^2 = 0.93$, $R_{MOS}^2 = 0.90$), forest cover ($R_{AEF}^2 = 0.94$, $R_{MOS}^2 = 0.89$), building footprint area ($R_{AEF}^2 = 0.77$, $R_{MOS}^2 = 0.78$), gross national income per capita ($R_{AEF}^2 = 0.70$, $R_{MOS}^2 = 0.72$), road length ($R_{AEF}^2 = 0.67$, $R_{MOS}^2 = 0.65$), global irrigated area ($R_{AEF}^2 = 0.59$, $R_{MOS}^2 = 0.52$), US tree cropped area ($R_{AEF}^2 = 0.38$, $R_{MOS}^2 = 0.16$), crude oil pipelines ($R_{AEF}^2 = 0.01$, $R_{MOS}^2 = 0.00$). The strong similarity in performance between models using AEF and MOSAIKS features supports the generalizability of our results, although additional comparisons in future work would be needed to fully understand how performance patterns compare between these and related approaches.

Third, like most machine learning based outputs, there is no direct physical interpretation for *how* MOSAIKS imagery features can predict any given variable. This contrasts with widely used metrics based on physical principles, such as the Normalized Difference Vegetation Index, which are designed to exploit known optical properties of target variables. Instead, MOSAIKS has a transparent mathematical explanation: it predicts values in new areas based on known values from areas that have similar image color, texture, and structure. This allows MOSAIKS to predict variables, such as educational attainment or literacy, for which there are no known spectral indices. It also helps explain why performance tends to decline as predictions are made farther from the training data, as places that look similar in the training data become less representative of the new area of interest.

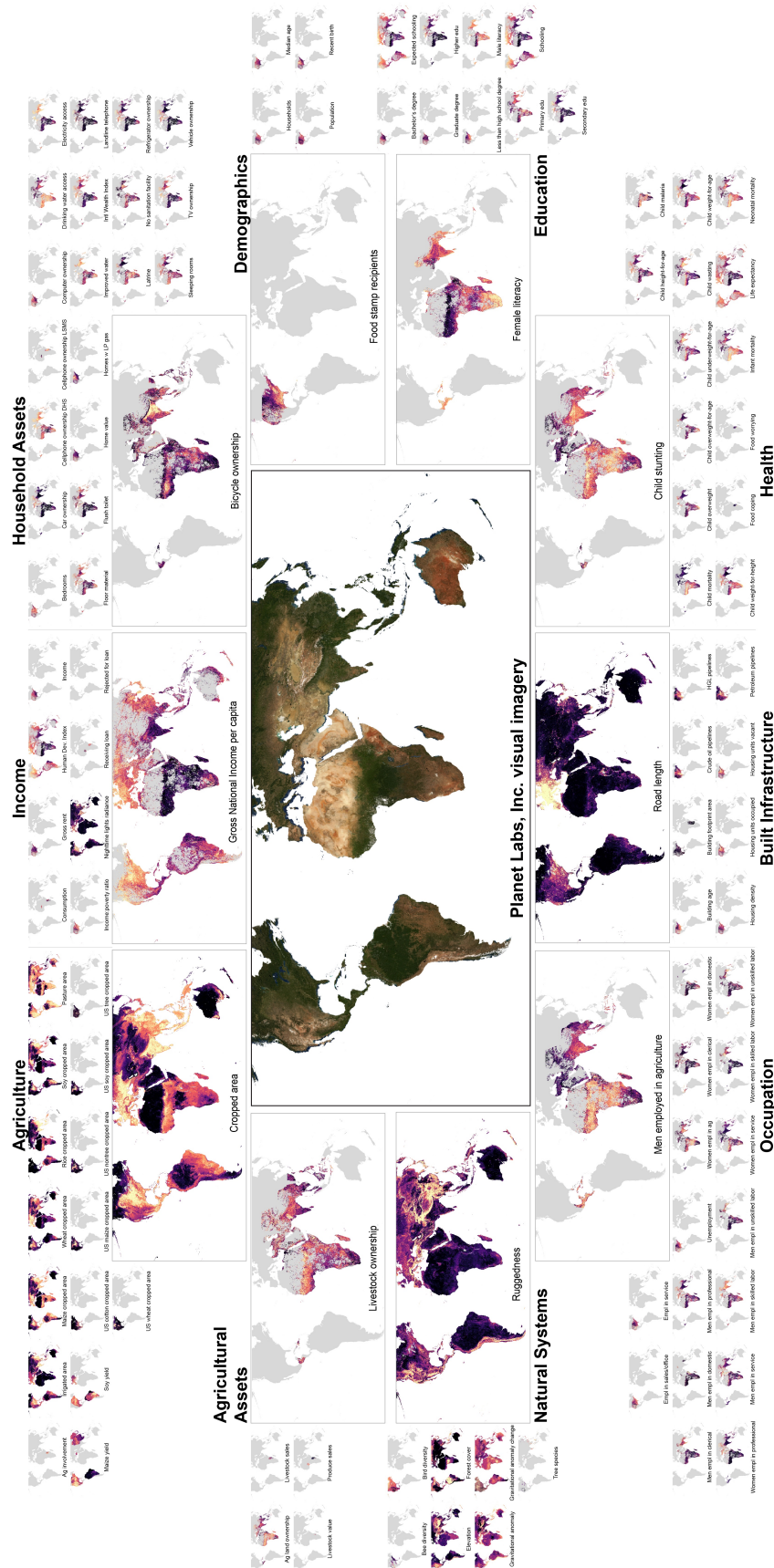
Fourth, the 115 variables we predict are not fully independent of one another and thus do not represent 115 fully orthogonal prediction tasks. For example, if income is easily predicted

from imagery and is highly correlated with other variables, our estimates of performance for these other variables may simply reflect the ability of imagery to predict income. We show that this possibility is unlikely to be a dominant driver of our recovered patterns of predictive performance with two diagnostic tests. First, Figure SM5 displays the correlation between all variables that spatially overlap in our ground truth dataset. While correlations are sometimes present, they are both positive and negative, and many variable combinations are uncorrelated. In a second test, we assess whether flexible functions of ground-measured income or wealth – principal axes of socioeconomic variation – predict more or less variation in ground conditions than our imagery-based models. Analyzing the 69 variables from the American Community Survey and the Demographic and Health Surveys, for which we have co-located measurements of income or wealth and other variables, we find that imagery-based models explain more variation in ground conditions than do income or wealth-based models for 60 variables, with an average R^2 difference of 0.20 (min = -0.14 for TV ownership and max = 0.88 for households per square km; see Figure SM6). This finding indicates that imagery-based models are leveraging information within the imagery that is distinct from and additional to income or wealth.

We hope this analysis helps guide the development and deployment of future SIML technologies. There has been both substantial optimism (1) and skepticism (21) regarding the potential of SIML-based monitoring in practice. Our findings indicate that there is rationale for both points of view: many variables can be estimated with high accuracy using even low-cost SIML technologies, while other ground conditions are not as easily predicted. Our broad set of experiments can help direct investment toward variables that are socially important and relatively easy to predict, particularly those that have thus far received little study. These experiments also uncover variables and settings for which development of SIML-based prediction systems may prove difficult or even impossible, highlighting the continued importance of traditional data collection methods.

Variable	R^2	Variable	R^2
Agricultural Assets			
Ag land ownership (% HH) (30)	0.58	Food coping (index) (29)	0.43
Livestock ownership (% HH) (30)	0.61	Food worrying (% HH in past wk) (29)	0.49
Livestock sales (\$/yr/HH) (29)	0.30	Infant mortality (deaths $\leq$ 1 yr/1000 live births (30)	0.12
Livestock value (\$/HH) (29)	0.28	Life expectancy (yrs) (31)	0.72
Produce sales (\$/yr/HH) (29)	0.01	Neonatal mortality (deaths $\leq$ 1 mo/1000 live births (30)	0.07
Agriculture		Household Assets	
Ag involvement (% HH) (29)	0.62	Bedrooms (per HH) (32)	0.48
Global cropped area (%) (33)	0.70	Bike ownership (% HH) (30)	0.68
Global irrigated area (%) (34)	0.52	Car ownership (% HH) (30)	0.53
Global maize cropped area (%) (33)	0.62	Cellphone ownership DHS (% HH) (30)	0.51
Global pasture area (%) (33)	0.64	Cellphone ownership LSMS (% HH) (29)	0.36
Global rice cropped area (%) (33)	0.63	Computer ownership (% pop) (32)	0.32
Global soy cropped area (%) (33)	0.51	Drinking water access (avg mins) (30)	0.57
Global wheat cropped area (%) (33)	0.60	Electricity access (% HH) (30)	0.67
Maize yield (kg/ha) (35)	0.90	Floor material (index) (30)	0.64
Soy yield (kg/ha) (35)	0.80	Flush toilet (% HH) (30)	0.65
US cotton cropped area (%) (36)	0.31	Home value (\$) (32)	0.62
US maize cropped area (%) (36)	0.66	Homes w LP gas (%) (32)	0.63
US nontree cropped area (%) (36)	0.78	Improved water source (% HH) (30)	0.36
US soy cropped area (%) (36)	0.70	Intl Wealth Index (30)	0.74
US tree cropped area (%) (36)	0.16	Landline telephone (% HH) (30)	0.52
US wheat cropped area (%) (36)	0.47	Latrine (% HH) (30)	0.66
Built Infrastructure		No sanitation facility (% HH) (30)	0.62
Building age (median yr built) (32)	0.59	Refrigerator ownership (% HH) (30)	0.72
Building footprint area (%) (37)	0.78	Sleeping rooms (per HH) (30)	0.33
Crude oil pipelines (km/km ²) (38)	0.00	TV ownership (% HH) (30)	0.68
HGL pipelines (km/km ²) (38)	0.00	Vehicle ownership (per HH) (32)	0.66
Housing density (units/km ²) (32)	0.86	Income	
Housing units occupied (%) (32)	0.40	Consumption (\$/pop) (29)	0.54
Housing units vacant (%) (32)	0.40	Gross Natl Income (\$/pop) (31)	0.72
Petroleum pipelines (km/km ²) (38)	0.00	Gross rent (\$) (32)	0.53
Road length (km/km ²) (39)	0.65	Human Dev Index (31)	0.78
Demographics		Income (\$/pop) (32)	0.49
Food stamps (% HH) (32)	0.43	Income:poverty ratio (32)	0.21
Households (HH/km ²) (32)	0.89	Nighttime lights radiance (nW/cm ² /sr) (40)	0.52
Median age (yrs) (32)	0.34	Receiving loan (% HH) (29)	0.20
Population (pop/km ²) (32)	0.88	Rejected for loan (% HH) (29)	0.04
Recent birth (% women) (32)	0.03	Natural Systems	
Education		Bee diversity (species) (41)	0.06
Bachelors degree (% pop $\geq$ 25 yrs) (32)	0.49	Bird diversity (species) (42)	0.55
Expected schooling (yrs) (31)	0.65	Elevation (m) (43)	0.62
Female literacy (% pop) (30)	0.55	Forest cover (%) (2)	0.89
Graduate degree (% pop $\geq$ 25 yrs) (32)	0.49	Gravitational anomaly (milligal) (44)	0.33
Higher edu (% of pop $\geq$ 6 yrs) (30)	0.44	Gravitational anomaly change (cm ewh/yr) (45)	0.62
Less than high school degree (% pop $\geq$ 25 yrs) (32)	0.43	Ruggedness (m) (46)	0.53
Male literacy (% pop) (30)	0.40	Tree diversity (species) (47)	0.74
Primary edu (% of pop $\geq$ 6 yrs) (30)	0.42	Occupation	
Schooling (yrs) (31)	0.72	Empl in sales/office (%) (32)	0.08
Secondary edu (% of pop $\geq$ 6 yrs) (30)	0.52	Empl in service (%) (32)	0.30
Health		Men empl in ag (%) (30)	0.45
Child height-for-age (z-score) (30)	0.20	Men empl in clerical (%) (30)	0.04
Child malaria (%) (30)	0.41	Men empl in domestic (%) (30)	0.22
Child mortality (deaths $\leq$ 5 yr/1000 live births (30)	0.22	Men empl in professional (%) (30)	0.13
Child overweight (% pop $\leq$ 5 yr) (30)	0.08	Men empl in service (%) (30)	0.19
Child overweight-for-age (% pop $\leq$ 5 yr) (30)	0.03	Men empl in skilled labor (%) (30)	0.19
Child stunting (% pop $\leq$ 5 yr) (30)	0.20	Men empl in unskilled labor (%) (30)	0.13
Child underweight-for-age (% pop $\leq$ 5 yr) (30)	0.32	Unemployment (%) (32)	0.20
Child wasting (% pop $\leq$ 5 yr) (30)	0.26	Women empl in ag (%) (30)	0.47
Child weight-for-age (z-score) (30)	0.44	Women empl in clerical (%) (30)	0.21
Child weight-for-height (z-score) (30)	0.39	Women empl in domestic (%) (30)	0.27
Food coping (index) (29)	0.43	Women empl in professional (%) (30)	0.30
		Women empl in service (%) (30)	0.32
		Women empl in skilled labor (%) (30)	0.16
		Women empl in unskilled labor (%) (30)	0.27

Table 1: 115 variables mapped using satellite imagery. Variables were chosen to span a diversity of socioeconomic and environmental categories and because publicly available data were available for model training and evaluation. See Supplementary Materials Section D for details on data sources. R^2 is out-of-sample predictive performance comparing satellite-based predictions to “ground truth” measurements from traditional sources.



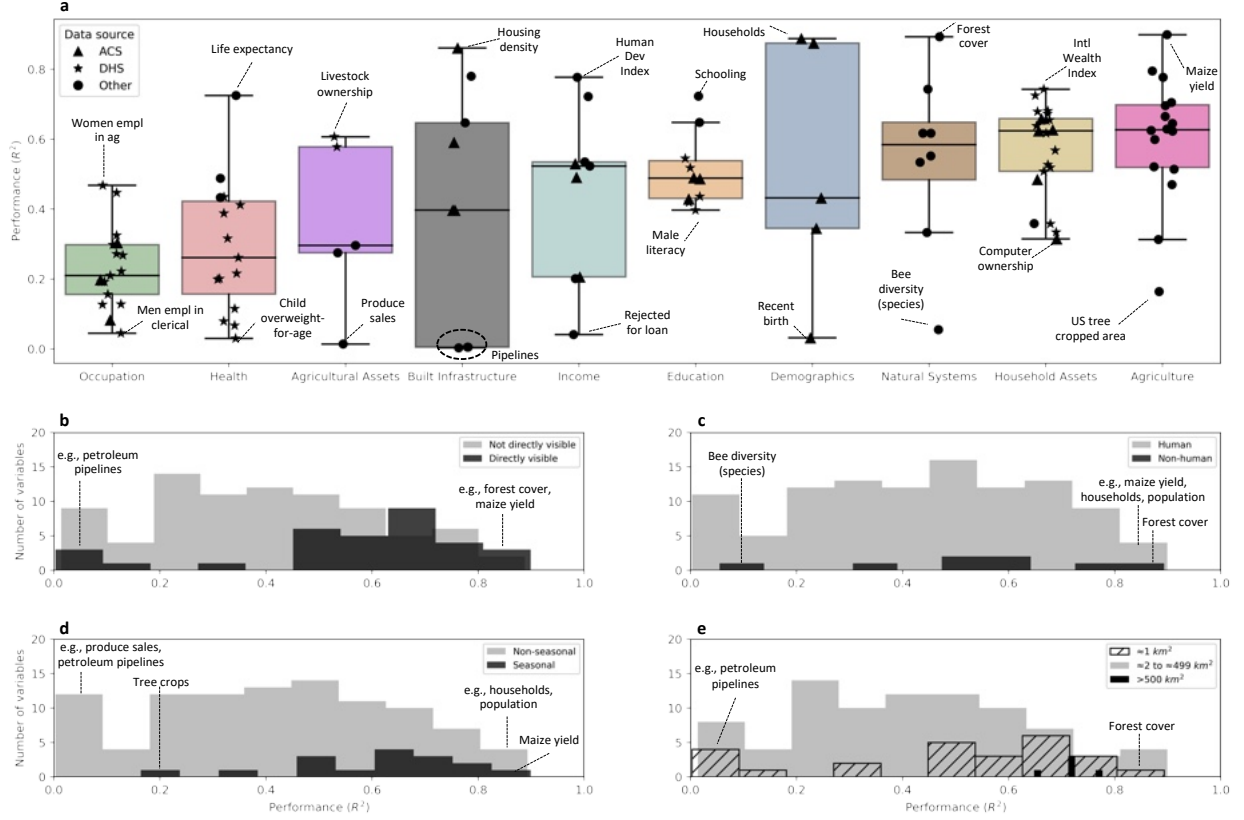


Figure 2: Performance of imagery-based mapping differs across variable categories and properties. All panels report out-of-sample performance (R^2) for 115 variables. Panel **a** shows the distribution of performance by category, ordered by mean performance. Performance for each variable is shown as a point, and visually summarized by a boxplot with the black horizontal line showing the median, box edges showing the interquartile range (IQR), and whiskers showing the most extreme data point within 1.5 times the IQR from the box edges. Panel **b** shows histograms of performance for variables that are either directly visible in imagery (light grey; e.g., forest cover) or only indirectly visible in imagery (black; e.g., years of schooling). Panels **c-e** show analogous histograms based on: **c** whether the variable represents a phenomenon produced primarily via human or non-human activity; **d** whether the variable represents a non-seasonal or seasonal phenomenon; and **e** the average native physical size of the ground truth measurements associated with each variable. Selected variables are identified in each panel to highlight examples of both high- and low-performing variables. Individual variable categories and properties are listed in Table SM2.

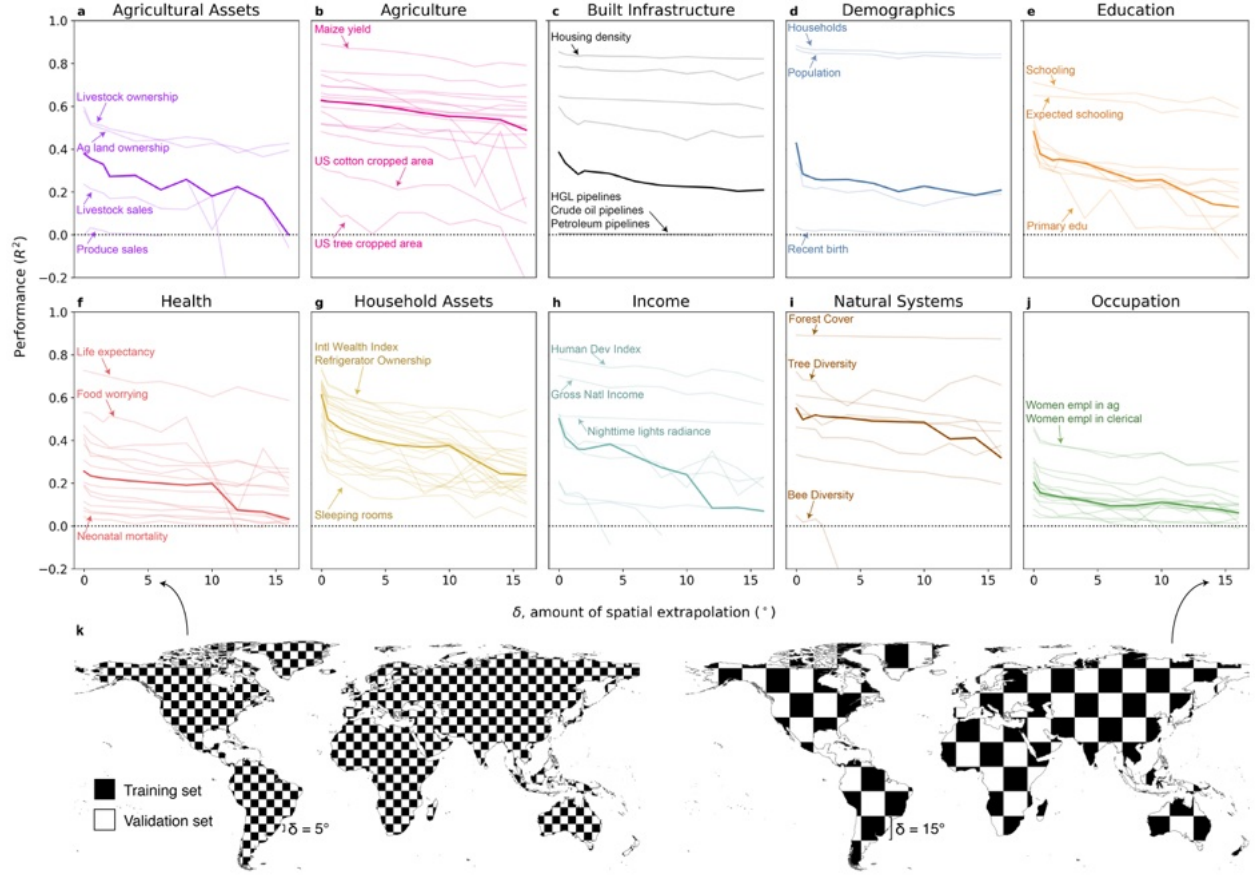


Figure 3: Model performance far from training data. Panels a-j show predictive performance in an experiment in which increasingly large physical distances separate training data from validation data. R^2 is shown on the y -axis. The x -axis is δ , a measure of spatial extrapolation. Specifically, δ is the width and height of grid cells (in $^\circ$) for a checkerboard pattern used to separate training data (black squares) and validation data (white squares), as shown in panel k for two example δ values (see Materials and Methods Section A.5 for details). In panels a-j, variables are grouped by category and each thin line corresponds to experimental results for one of the 115 variables investigated in this study. Median performance across variables in each category is plotted with a darker line. The dotted horizontal line shows where $R^2 = 0$. Example variables are labeled to highlight cases of both high and low performance.

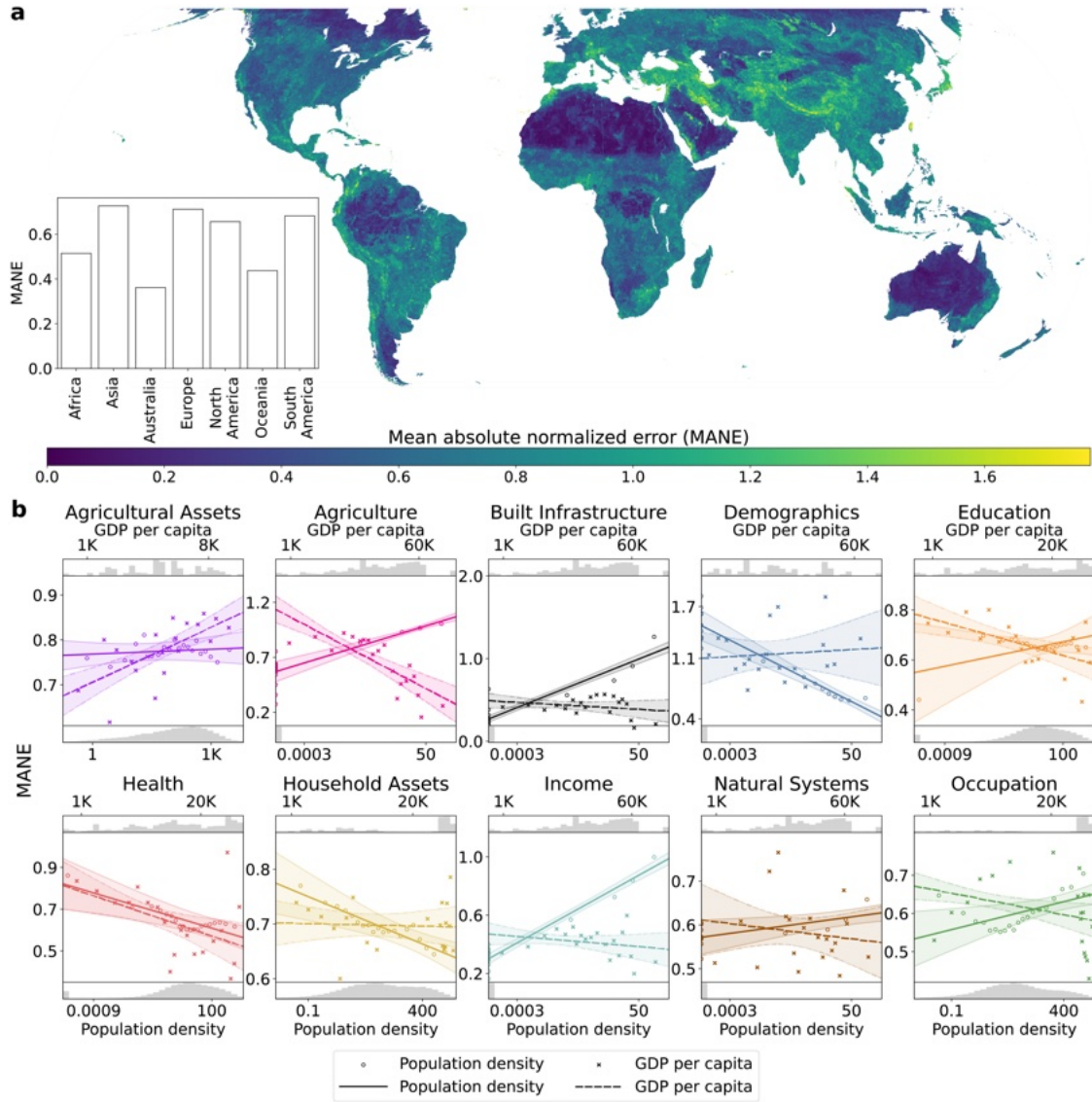


Figure 4: Patterns in map accuracy across many maps. Panel **a** maps Mean Absolute Normalized Error (MANE) at $0.01^\circ \times 0.01^\circ$ resolution (Supplementary Materials Section A.5). For each location, MANE quantifies an average measure *across all variables* of standardized accuracy *within each variable*. The value of MANE in a location indicates the average normalized (i.e., $\mu = 0, \sigma = 1$) prediction error across all variables with available ground truth data in that location (yellow indicates higher rates of average prediction error). The inset boxplot shows average MANE by continent. Panel **b** shows how MANE varies by levels of GDP per capita (top axes) and population density (bottom axes). In each sub-panel, relationships between MANE and GDP per capita are indicated by dashed regression lines and crosses, while relationships between MANE and population density are indicated by solid regression lines and circles. Data points shown (i.e., crosses and circles) are binned scatterplots using 20 bins. 95% confidence intervals are shaded and are estimated allowing for arbitrary correlation in model errors within first-level administrative units. To remove the influence of outliers in **b**, the 1st and 99th percentiles of the population density and GDP per capita distributions are dropped for each category.

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Code and data availability Replication code will be made publicly available upon publication. All data used in this analysis, other than from the Demographic and Health Surveys and the Living Standards Measurement Surveys (Supplementary Materials Sections D.1 and D.20), is from free, publicly available sources and will be made publicly available upon publication. Maps and accompanying data produced in this analysis will be made freely available at <https://www.mosaiks.org> upon publication.

List of Supplementary Materials

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- Supplementary Text: Automated model specification
- Supplementary Text: Comparison to prior literature
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Supplementary Materials for

What can satellite imagery and machine learning measure?

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A Materials and Methods

A.1 Experimental design

What can and cannot be accurately predicted using satellite imagery is difficult to test empirically. An ideal experiment would train and evaluate models for all relevant social and environmental variables, using an algorithm that captures all predictive imagery information and ground truth measurements that are error-free, identically structured, and evenly distributed. In practice, we approximate such an experiment by analyzing a diverse set of 115 variables using a standardized remote sensing pipeline that performs competitively with other leading approaches on a wide variety of tasks. We facilitate comparisons across variables and locations by using a single imagery source and machine learning method. Our analysis relies on high-resolution optical imagery, due to its widespread use and its demonstrated ability to capture textural information relevant to both socioeconomic and environmental variables. Using a large sample of variables and a single remote sensing methodology, we aim to provide practitioners with useful guidelines on the opportunities and limitations of using high-resolution optical imagery for mapping ground conditions.

A.2 Ground truth data

We collected ground truth data for 115 variables across a variety of spatial scales and scopes. These variables were selected to span a wide range of social and environmental outcomes that are central to human and environmental well-being. Selection was also determined by the availability of data; importantly, we limited our collection efforts to publicly available sources. Most variables derive from surveys, censuses, or other forms of direct measurement. Others are remotely sensed outcomes that were estimated using different satellites and methodologies than the approach taken here (e.g., we predict nighttime luminosity using daytime satellite imagery). In some cases, we include the same variable from multiple sources (e.g., survey-based assessments of cellphone ownership) in order to understand the potential impacts of data source and/or spatial scope on model performance. Predictive performance for all variables that were analyzed is reported in the main text.

All collected data must be processed and matched with satellite image features before it can be used for model training and evaluation. We process data to fit into two structures, based on the native form of the ground truth data: “polygon” variables, and “grid” variables. For polygon variables, each observation is composed of a value for that spatial polygon (e.g., the number of households per km² in a census tract from the American Community Survey (ACS)) and a vector of satellite features for that spatial polygon (the processing of which

is discussed in the following section). For grid variables, each observation is composed of a value for a $0.01^\circ \times 0.01^\circ$ grid cell (e.g., the fraction of the grid cell covered in forest) and a vector of satellite features for that grid cell. To reduce both spatial correlation between observations and computation, we process and analyze grid observations on a sub-sampled “sparse grid” rather than all grid cells. The sparse grid is a 1% sample of the full $0.01^\circ \times 0.01^\circ$ grid, created by taking every 10th cell in the latitude and longitude directions.

Whether each variable is modeled as a polygon or grid is indicated in Table SM2. All Demographic and Health Surveys (DHS) variables, all ACS variables, and all variables whose ground truth values are provided in raster format at coarser than $0.1^\circ \times 0.1^\circ$ resolution are modeled as polygons. All other variables are modeled as grid-level observations. For grid variables where the ground truth values have a native resolution that is coarser resolution than the grid cells (i.e., $0.01^\circ < \text{resolution} < 0.1^\circ$), we assign the grid cell the value of the coarser measurement with the closest centroid. For grid variables where the ground truth values have a native resolution that is finer than 0.01° , we aggregate values to the $0.01^\circ \times 0.01^\circ$ grid. Processing steps for all variables are detailed in Supplementary Section D.

When possible, we construct all variables to have intensive rather than extensive values (e.g., computing county average crop yields rather than total county crop production), to ensure that variable values are not a direct function of the size of the variable area. Using variables with intensive values, imagery features that are averages over the variable area (as detailed in Section A.3), and a linear prediction model, collectively allow for the use of variables of varying shapes and sizes in model training and evaluation (19).

A.3 Satellite image features

We compile a high-resolution global cross-sectional composite of visual imagery from Planet Lab’s Surface Reflectance Basemaps product from the third quarter of 2019. This product has a pixel resolution of $4.77\text{m} \times 4.77\text{m}$ at the equator. Planet Labs processes these quarterly mosaics to minimize cloud cover, balance color across seasons, and remove seams from images (16, 49). We use data from the third quarter of the year to minimize both northern hemisphere ice coverage and tropical cloud cover.

Using this imagery, we calculate 4,000 task-agnostic imagery “features” using random convolutional features (RCF) (7, 50), the implementation for which is detailed in ref. (7) and on our public website (49). These features translate raw imagery into a set of tabular data, encoding rich information about each satellite image’s colors, patterns, and textures into a vector of feature observations per image. We generate RCF imagery features for all global

locations with available imagery on a global $0.01^\circ \times 0.01^\circ$ equal-angle grid. Imagery coverage is comprehensive beneath 60° latitude, after which images and associated features are often missing. 2,000 of these imagery features are constructed using a patch size of $4 \times 4 \times 3$ pixels, and the other 2,000 use a patch size of $6 \times 6 \times 3$ pixels, where the third dimension represents the red, blue, and green visual bands and the first two dimensions represent the width and height of each patch (see ref. (7) for details). This combination of patch sizes was found to optimize performance on three test variables: nightlight luminosity, road length, and forest cover, when patch sizes ranging from $3 \times 3 \times 3$ to $10 \times 10 \times 3$ were tested. This process produces more than 440 billion imagery feature values. In evaluating the quality of these features, we find that $> 99.5\%$ of them appear to be internally consistent. A small fraction of the features exhibit visible discontinuities at the borders of the pre-processed, analysis-ready Planet Labs imagery we use as inputs. These discontinuities may be an artifact of the underlying imagery or our feature production pipeline. We compare model performance across a variety of tasks in areas more vs. less affected by the discontinuities and do not find meaningful differences.

We spatially match ground truth data for each variable to its respective image feature vector. For grid variables, we use the features for the respective grid cell. For polygon variables, we average grid cell feature values across all grid cells overlapping each polygon. Averages over polygons are calculated weighting by either area or by population; we evaluate the performance of both approaches for each variable and select the method that performs best, as described in the following section. To compute population weights, we use population data from the Gridded Population of the World V4 (GPW), which has 0.008° resolution (51). For some polygon variables with large extents we aggregate image features from a 1% sample of the total area to reduce computation; these include maize yield, soy yield, gravitational anomaly change, expected schooling, gross national income per capita, human development index, life expectancy and years of schooling. For all other polygon variables, weighted average feature vectors are computed using the full 0.01° resolution grid.

A.4 Model training

With a merged set of ground truth data and tabular feature data for each of our 115 outcome variables, we train cross-validated ridge regression models to learn the relationship between imagery features and each outcome variable of interest. Specifically, we randomly split each dataset of merged ground truth and feature values into a training/validation set (80% of the sample) and a test set (20% of the sample). We train linear ridge regression models in the training/validation set using 5-fold cross-validation to tune model hyperparameters (i.e., the ridge penalization parameter). After training, model performance is evaluated and reported

using the held-out test set. Performance is similar when evaluated in the validation and test sets, indicating that the model was not over-fit to the training data (Figure SM7). Throughout this process, data are randomly partitioned. The only exception to random partitioning is in the experiment shown in Figures 3 and SM3, where splits between training/validation and test sets are chosen specifically to ensure differing degrees of spatial extrapolation between the two sets. Since performance evaluation using cross-validation should be designed to match the way in which the model is used and its performance interpreted (52), we use random splits of the data to quantify the ability of the models to interpolate missing data drawn from a similar distribution as the training data. For example, if one were to collect maize yield data from a randomly selected 80% of sub-national administrative units across the U.S., China, Europe and Brazil, this approach tests the ability to predict the remaining 20% of unmeasured values. Whether satellite-based models can generalize to areas systematically distant or distinct from the areas in which training occurred is another important question, which we evaluate using the spatial cross-validation approach. This mirrors a situation where, for example, yield values are measured in large contiguous groups of administrative units, but not in others, and the model is used to predict values in the unmeasured regions.

To facilitate model training and evaluation at scale, we standardize and automate each step of the model training pipeline. Details of our approach to automation are provided in Supplementary Materials Section E, but we summarize the method here. We tune all model hyperparameters, such as the degree of ridge regression penalization, as well as more complex decisions, such as whether to model the variable in logarithms or levels, by training all combinations of possible parameter choices and selecting the model that performs best in 5 held-out validation sets. We then retrain the model using the selected hyperparameters on the full training/validation set and evaluate performance in the test set. Model tuning decisions automated in this way include: whether to allow different imagery prediction models in different regions of the globe or to use a single global regression model; whether or not to winzorise predictions to the training sample range; whether to model the outcome in levels or in logarithms;¹ whether or not to include a model intercept; and whether to weight by population or by area when aggregating features to the polygon-level. We find that such tuning, which is made feasible by the computational efficiency of linear ridge regression using unsupervised features, improves model R^2 on average by 0.03 ($\sigma = 0.03$, greatest improvement = 0.18), relative to a baseline model that optimizes the ridge regression penalization term and log/level decision separately for each variable, but then uses a

¹Because logarithms are more difficult to interpret than variables reported in levels, we choose to model the variable in log form only if the logarithmic outcome out-performs the levels model by at least 10%.

spatially-homogeneous mapping between imagery and the outcome for all cross-continental variables, uses an intercept in all models, and aggregates all feature values to polygons using area weights. Importantly, this pipeline enables any new prediction to be generated quickly for any new variable from available ground truth data with very limited user input. In other contexts, these modelling decisions may be customized to meet the specific needs of the task (e.g., predictions may be desired in levels or in logs). Because we focus here on developing a pipeline that scales across variables, we use the model that performs best in this automated procedure for standardization and simplicity.

A.5 Analysis of performance

We report model performance as compared to prior studies (Figure SM1), across variable types (Figure 2), across space and socio-economic characteristics (Fig. 4), and at varying degrees of spatial extrapolation (Figure 3). Each of these analyses are detailed in the following subsections.

Comparison to prior studies To compare performance of the pipeline developed here with that of the previous literature (Figure SM1), we conducted a review of previous analyses using satellite imagery to predict socio-economic and environmental conditions. For each of the 115 variables we analyzed, we searched for published papers measuring the same (or a very similar) variable and recorded their performance, as well as other metadata to facilitate comparison between our study and prior analyses. The results from this review are shown in Table SMC. To restrict the scope of this review and ensure comparability to our analysis, we evaluated only papers that: generated predictions at the scale of a single country or larger (though we made an exception for subnational studies with an exceptionally wide geographic scope, e.g., (12)); used at least one predictor variable derived from satellite imagery; and were published in a peer-reviewed journal at the time of the literature review (January 2024). When a single paper reported multiple performance values, we report the average of those values. When multiple previous papers exist for the same variable, we compared our performance to the paper with the spatial extent closest to that of our predictions.

This comparison provides a general assessment of how the performance of our imagery-based pipeline compares with that of previous literature. However, direct comparison of performance across studies is complicated by many factors. First, for prediction of continuous variables, studies report model performance using different metrics, including: the coefficient of determination (R^2 , used throughout our analysis), the square of the Pearson’s correlation coefficient (r^2), and the adjusted coefficient of determination (R^2_{adj}). Further, some studies don’t specify the precise metric used. As an example of how these differences complicate

comparison, r^2 performance is invariant to a mean shift and to a re-scaling, while R^2 will decline if predictions are offset from ground truth either by a mean shift or by a rescaling factor. This makes r^2 performance systematically higher than R^2 . Second, some prior analyses use other covariates alongside satellite features in their predictive models, enhancing predictive performance. For instance, ref. (53) use a discrete hazards geostatistical model with nighttime lights, air pollution, and other non-satellite geospatial covariates to map neonatal, infant, and child mortality across all low- and middle-income countries. Similarly, ref. (54) use satellite-derived vegetation indices along with a suite of climate and other environmental variables in a linear model to predict global patterns of bee species richness. As our study uses only satellite features to predict all outcomes, its performance can be viewed as a baseline, which could potentially be improved by the addition of other covariates into the linear regression modeling step (7). Finally, other differences between our analysis and previous studies, such as spatial and/or temporal resolution, whether predictions are generated from country-specific or pooled multi-country models, and the precise definition and data source of the outcome variable, can also complicate direct comparison. Where possible, we have indicated key differences between our analysis and other studies in Figure SM1 using colors and symbols, but such coding remains imperfect given the many dimensions along which studies can differ.

Comparison across variable types To compare performance across variable types (Figure 2), variables were grouped into categories (e.g., built infrastructure, demographics, occupation, etc.). We chose categories to represent conceptual groupings of the 115 variables under the constraint that each category contained at least five variables. Categories were constructed and variables were assigned prior to assessment of model performance. Other variable characteristics shown in Figure 2, such as whether the variable is directly visible or not and whether it is human or non-human, were also decided by author consensus without consideration of model performance.

Comparison across space and socio-economic characteristics To analyze how predictive performance varies across locations and socio-economic characteristics (Figure 4), we develop a measure of model error that is designed for comparisons across space. We call this metric the Mean Absolute Normalized Error metric, or “MANE.” MANE is computed for each $0.01^\circ \times 0.01^\circ$ grid cell as the average of the absolute value of error Z-scores, where the average is taken over all variables for which ground truth data are available in that grid cell. MANE is thus a measure of average performance in each location: a higher value of MANE means that, on average, our predictions are further from their ground truth value

in that place. To construct MANE, we first calculate prediction errors for each variable in each $0.01^\circ \times 0.01^\circ$ grid cell for which ground truth data are available. We then standardize these errors to ensure their distributions are mean zero and standard deviation one for each variable, and take their absolute value to get absolute normalized error for each variable. Locations with predictions that match ground truth data better than the rest of the locations have lower absolute normalized error, while locations where predictions tend to be different from ground truth data values have larger absolute normalized error. We then average these absolute normalized errors across all variables available in each location resulting in a single MANE value for each grid cell.

Mathematically, MANE is calculated as:

$$\text{MANE}_i = \frac{1}{|L_i|} \sum_{l \in L_i} \left| \frac{\hat{\varepsilon}_{li} - \bar{\varepsilon}_l}{\sigma_l} \right|,$$

where i indexes the grid cell (i.e., location), l indexes the variable, L_i is the set of variables for which ground truth data are available at grid cell i , $\hat{\varepsilon}_{li}$ is the prediction residual for variable l at grid cell i , $\bar{\varepsilon}_l$ is the mean prediction residual for variable l across all grid cells with available ground truth data, and σ_l is the standard deviation of the residuals for variable l across all grid cells with available ground truth data. For predictions made at the polygon level, we rasterize the polygons to the $0.01^\circ \times 0.01^\circ$ grid before calculating MANE.

In Figure 4A, we map MANE at the grid cell level to visualize average performance across space. In Figure 4B, we merge MANE with grid cell level estimates of population density from the Global Human Settlements Layer (48) and with GDP per capita estimates from ref. (55) to quantify how performance varies across these dimensions.

Spatial extrapolation Prior literature has established that imagery-based predictions are less reliable estimates of ground truth when they are made spatially far from available training data (7, 52). To evaluate the ability of our imagery-based model to make predictions in areas spatially distant from training data, we conduct an experiment in which we split the data from the training and validation data sets into a checkerboard pattern, giving two disjoint sets of roughly equal size. We train the model on data from the black squares of the checkerboard pattern and evaluate model performance on the white squares (for a visual illustration of this approach, see Figure SM3). Increasing the size of the checkerboard forces increasingly large spatial separation between the datasets used for model training versus those used for evaluation of performance. We use checkerboards with widths and heights of 0.5, 1.5, 2.0, 4.0, 6.0, 8.0, 10.0, 12.0, 14.0, 16.0 degrees, or around 50km to 1750km. This experiment follows a similar procedure as in ref. (7) – see Supplementary Note 2.8 therein

for details. For variables represented as polygons in ground truth data, we consider a polygon to be in a given checkerboard grid cell based on the location of the polygon centroid. All model tuning is conducted in the training checkerboard data set before the model is applied to the checkerboard validation set.

Comparison to models using AlphaEarth Foundations features Models using AEF features were trained, tuned, and evaluated using the same approach as the MOSAIKS features. Like the MOSAIKS features, the AEF features were aggregated to $0.01^\circ \times 0.01^\circ$ resolution before being matched to the ground truth label data. To allow for a nonlinear relationship between the AEF features and the outcome, we calculate in each $0.01^\circ \times 0.01^\circ$ grid the fraction of each AEF feature value falling within one of ten quintile bins and use these fractions as features. We take this approach because it performs slightly better than using the mean of the AEF feature values in each grid. We evaluate performance for eight variables with ground truth values collected for a global sparse grid (Table SM2 and Methods A.2); this subset of variables was chosen because they share a common geographical range and thus could be analyzed efficiently, given the time required to download and process the full set of $10\text{m} \times 10\text{m}$ AEF features at global scale.

A.6 Production and distribution of maps

Global satellite-based predictions For each of the 115 variables analyzed here, we produce global-scale maps of imagery-based predictions at the scale of a $0.01^\circ \times 0.01^\circ$ grid. To do so, we combine trained models for each variable, the performance of which is shown throughout the main paper, with pre-computed random convolutional features for all grid cells over land area and with available imagery (see Section A.3 for details on our imagery, which is sourced from Planet Labs, Inc.). We distribute predictions only in areas with positive predictive performance and only in locations within 8° of training data (i.e. half of the maximum tested checkerboard width). For socio-demographic variables we distribute predictions only in areas that are settled by humans, as estimated by the Global Human Settlements Layer.

Data quality scores Alongside each of the 115 global-scale sets of imagery-based predictions, we provide a spatially explicit and variable-specific estimate of the data quality provided by these predictions. This data quality layer provides important information regarding the reliability of the imagery-based predictions. Specifically, we use the results from the spatial extrapolation experiment, described above, which estimates how predictive power degrades as a function of distance from the training data. From these results, we record an

estimate of predictive performance for each out-of-sample location and each variable. For example, if the spatial extrapolation experiment uncovered an R^2 of 0.3 when using a checkerboard of width 2.0° for a given variable, we would estimate performance of approximately $R^2 = 0.3$ for that variable for any grid cell located approximately 1.0° away from any available ground truth data. We assign performance recovered from a checkerboard width of δ degrees to a spatial extrapolation distance of $\frac{\delta}{2}$ because in this experiment, the maximum distance between a training observation and a test set location is $\frac{\delta}{2}$. To simplify the data quality reporting, we then categorize predictive performance into six quality scores: variable-locations with an $R^2 < 0$ receive a quality score of zero; those with $0 < R^2 \leq 0.15$ receive a score of 1; those with $0.15 < R^2 \leq 0.3$ receive a score of 2; those with $0.3 < R^2 \leq 0.45$ receive a score of 3; those with $0.45 < R^2 \leq 0.6$ receive a score of 4; and those with $R^2 \geq 0.6$ receive a score of 5. When regional models are used in place of a single global-scale imagery-based prediction model, this procedure of conducting the checkerboard experiment and generating data quality scores is applied separately to each region. We do not allow quality scores to increase with distance from the training observations, which happens in a few cases due to noise (for examples, see figures in Supplementary Materials Section F). Because the spatial extrapolation experiment evaluates predictive performance up to a spatial extrapolation distance of 16 degrees, we report a quality score of zero for predictions located farther than $\frac{\delta}{2} = 8^\circ$ from available training data.

We illustrate this procedure for a single variable and continent in Figure SM3. In this example, we consider the continent-specific model for Africa for the variable measuring the percent of households with a flush toilet. In this example, we achieve an $R^2 = 0.66$ when the checkerboard width is 0.5° (this is the smallest checkerboard width we evaluate). This result is shown in Fig SM3a. With a checkerboard of this size, training and validation data are on spatially separated by a maximum of 0.25° . Because this performance exceeds $R^2 = 0.6$, we apply a data quality score of 5 for all points in Africa that are within 0.25° of any existing DHS clusters in the full set of ground truth data (Figure SM3a). When we increase the checkerboard width up to 16 degrees (this is the largest checkerboard width we evaluate), out-of-sample performance declines to $R^2 = 0.3$ (Figure SM3c). Thus, we assign a data quality score of 3 for all locations between 6° and 8° away from a DHS cluster. As shown in Figure SM3d, the experiment implies a quality score of 4 for all other locations between 1° and 6° away from available ground truth for this variable. For locations more than 8° from ground truth data, the data quality score is assigned to be zero.

Data release The data product we distribute includes up to four layers, all of which are provided on a global grid with $0.01^\circ \times 0.01^\circ$ resolution. The first layer is a set of grid cell

predictions of 115 distinct variables. For socio-economic outcomes, these predictions are masked to only include locations where population is estimated to be nonzero; we use the Global Human Settlement Layer (48) to create this mask. The second data layer we release contains the 0-5 data quality scores, as described above. The third data layer contains the values of the ground truth data that we use for training, rasterized to the grid cell level from the native resolution of each underlying dataset as described above in Section A.2. The final layer is included only for polygon variables and describes which grid cells correspond to which polygons from the ground truth data. These final two layers are available only in the locations where we have obtained ground truth data. Note that we cannot release this data publicly for the variables derived from the DHS or from the Living Standards Measurement Survey (LSMS).

B Supplementary Figures

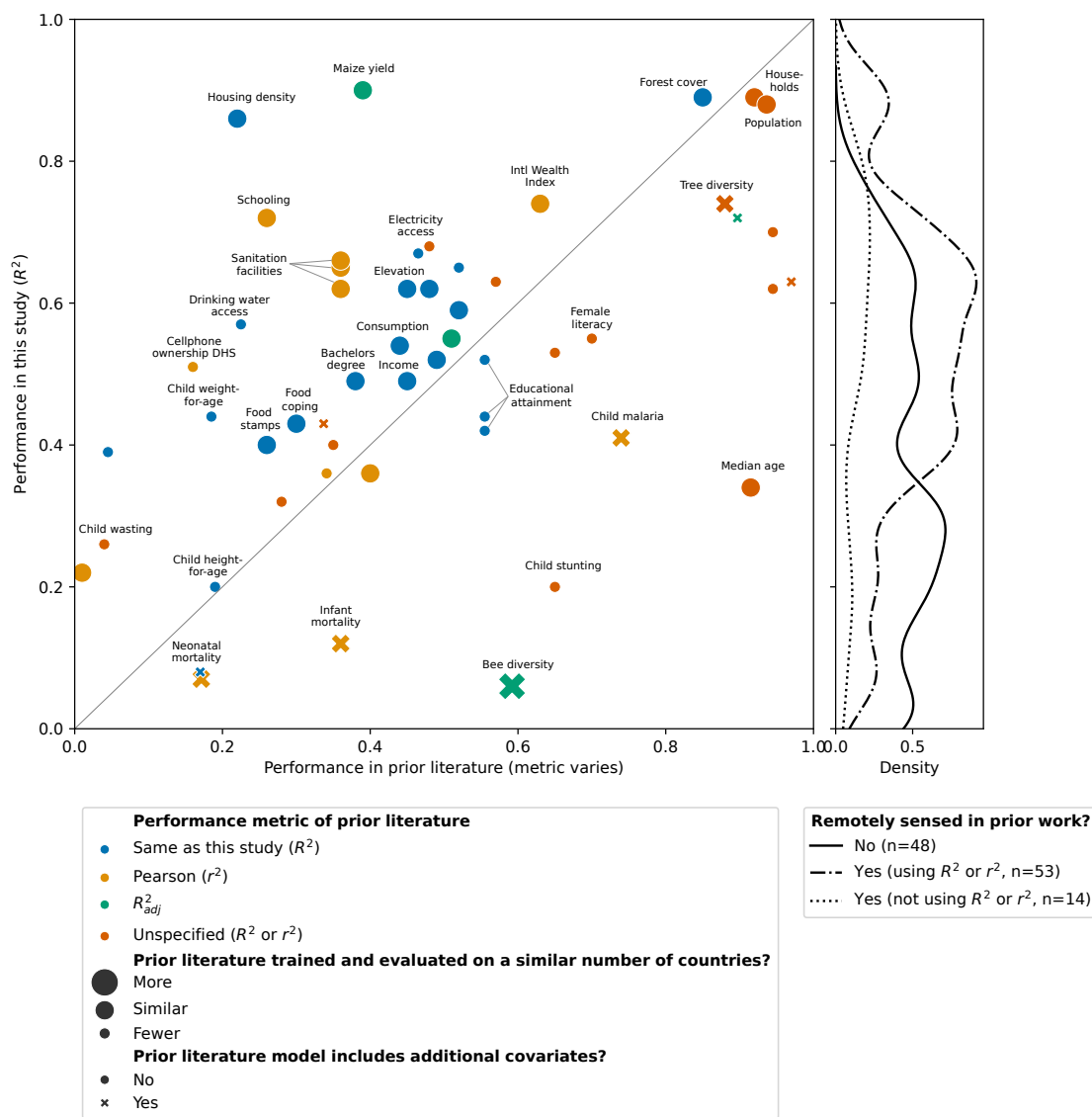


Figure SM1: Comparison of map accuracy between this study and prior work. Out-of-sample map accuracy (R^2) in this study compared with reported performance from prior studies that use satellite imagery to predict similar variables at the country scale or larger. Of the 115 variables mapped in this analysis, 67 have been mapped in previous studies, and, of these, 53 have been evaluated using R^2 or r^2 ; the performance of these is shown in the scatter plot (see Materials and Methods Section A.5 for details on this literature review). Symbol color indicates the metric used to evaluate performance in prior work, symbol size indicates the spatial scale of prior analysis relative to this study, and symbol marker indicates whether prior analysis included ancillary data other than satellite imagery in their predictive models (this analysis used no ancillary data). The 45° line indicating equal performance between this study and prior work is shown in light grey. Density plots on the right show the distribution of our model performance for variables that have been predicted using imagery in prior work (dotted and dashed lines) and those predicted for the first time in this study (solid line). In some cases, multiple of the variables we remotely sense correspond to a single previous estimate. For example, we compare our performance estimating flush toilets, latrines, and no sanitation facilities to previous performance estimating a sanitation index.

Observed



Predicted

Figure SM2: Imagery-based predictions compared with ground truth for 115 variables. Each plotted point represents one training observation; 45° lines are shown in grey. Performance metrics (r^2 and R^2) shown here are computed out-of-sample in the training/validation set. Test-set performance is shown in Table 1.

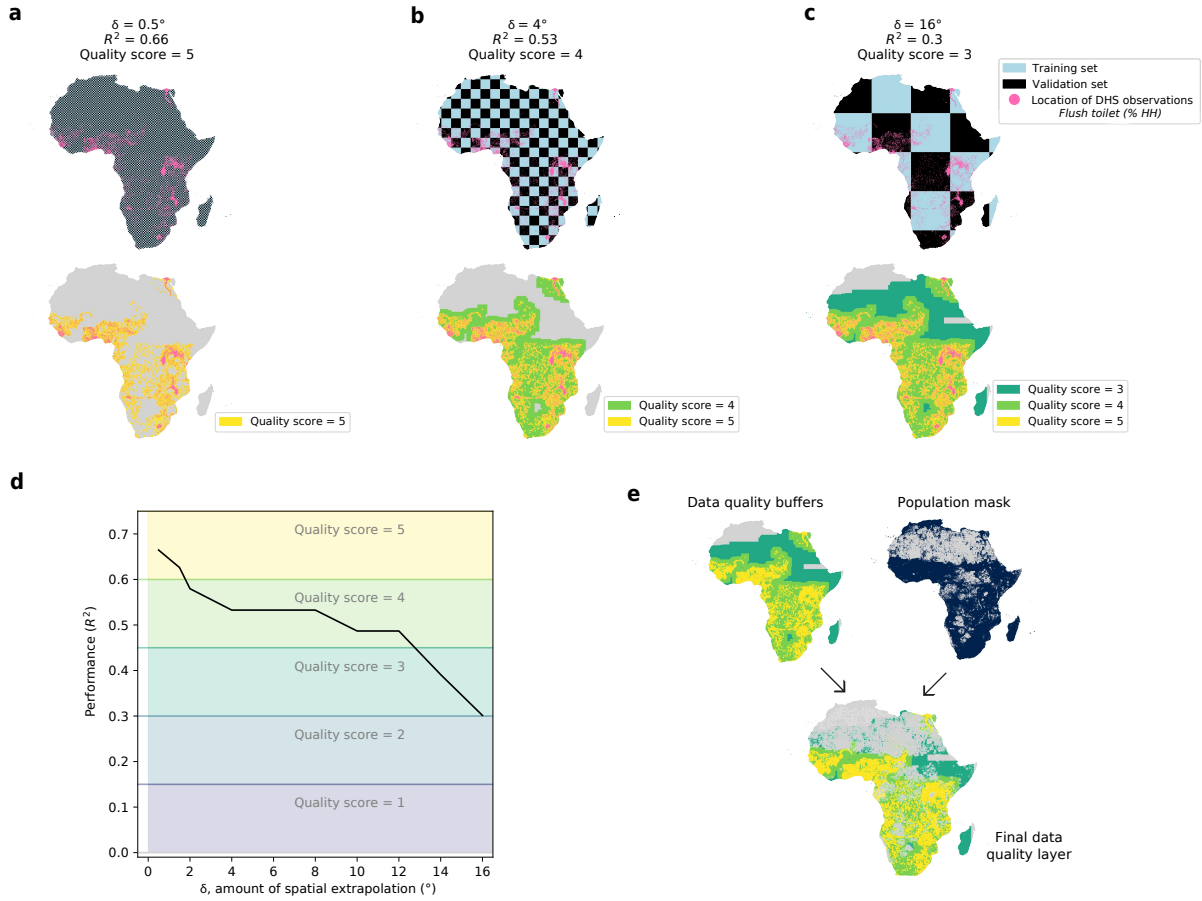


Figure SM3: Illustration of the approach used to generate spatially-explicit data quality scores from model performance in a spatial extrapolation experiment. Figure illustrates how data quality scores are generated for an example variable (percentage of households with a flush toilet) and region (African continent). Checkerboard patterns in panels **a-c** show the division of ground truth observations (pink) into a training sample (light blue squares) and a validation sample (black squares). The width and height of the checkerboard determines the extent of spatial separation between training and validation data and is indicated by δ , measured in degrees. Panel **d** shows performance for different values of δ , following Figure 3, and displays how performance in this experiment translates into quality scores on a 0-5 scale (where 0 indicates any location outside the spatial extent of the experiment or non-positive performance in the experiment). Colored maps in panels **a-c** show data quality buffers, indicating the quality score from panel **d** that applies to each location in the continent based on its distance from training data. Panel **e** shows how these data quality buffers are used in combination with a population mask derived from the Global Human Settlements Layer (48) to create a final data quality layer for all grid cells in Africa where population density is nonzero. See Materials and Methods Section A.6 for details on this approach.

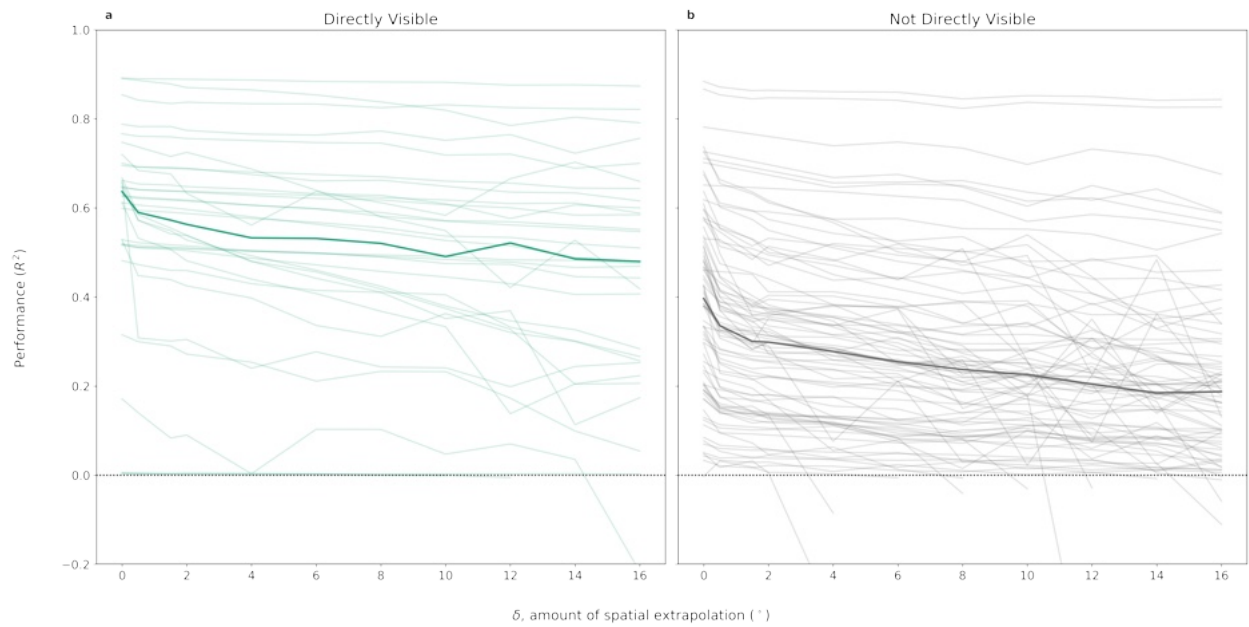


Figure SM4: Model performance far from training data for directly visible versus not directly visible variables. Identical to Figure 3, but variables are grouped by whether they are directly visible in the image (panel **a**) or not directly visible (panel **b**), as in Figure 2. A list of all variables and whether they are categorized as directly visible or not is provided in Table SM2.

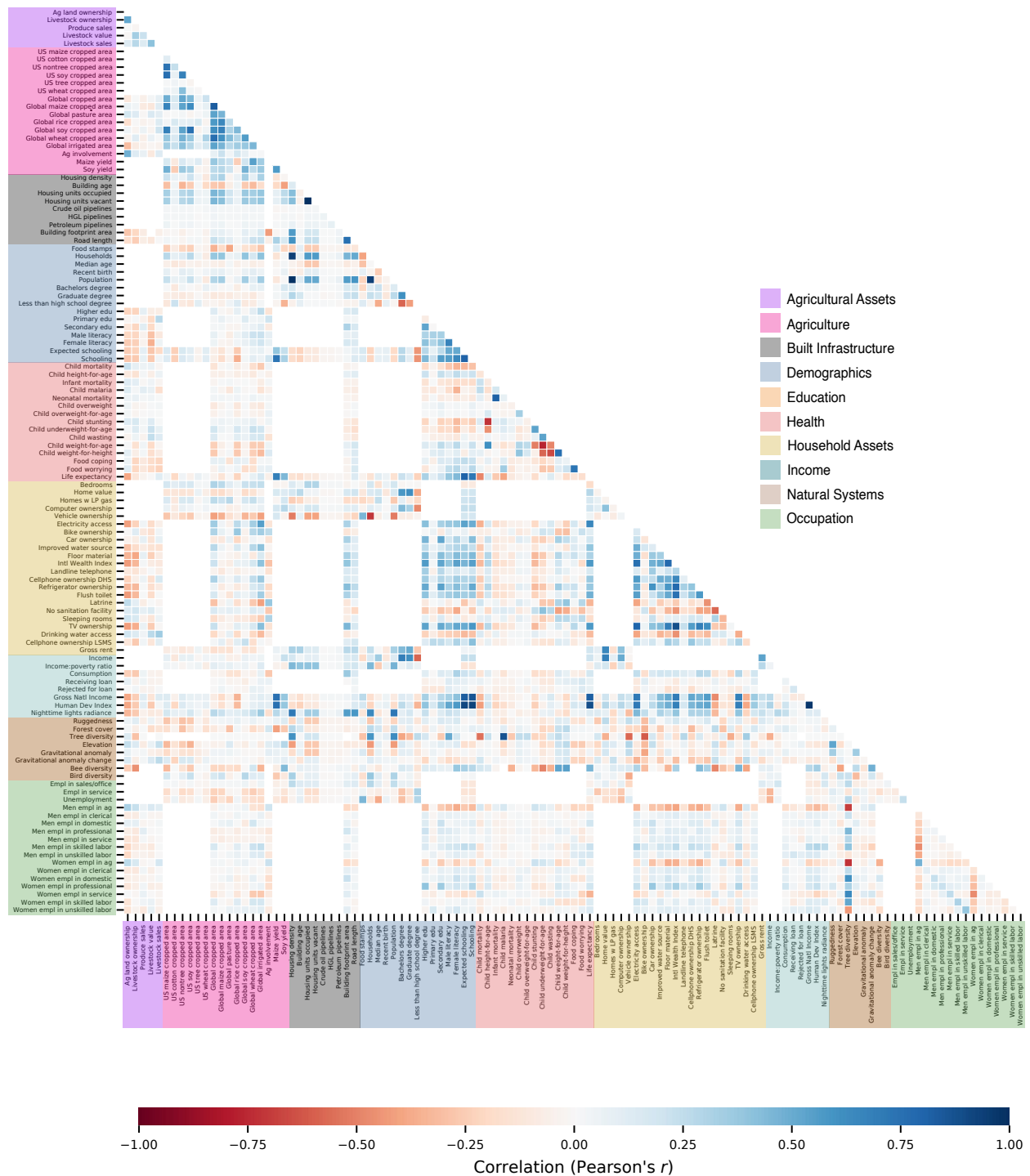


Figure SM5: Correlation matrix across 115 variables mapped in this study. Correlations are calculated from ground truth values, and are thus only available for pairs of variables with spatial overlap. Dark blue denotes high positive correlation and dark red denotes high negative correlation. Shading on the x - and y -axes correspond to the 10 variable categories shown in Figure 2.

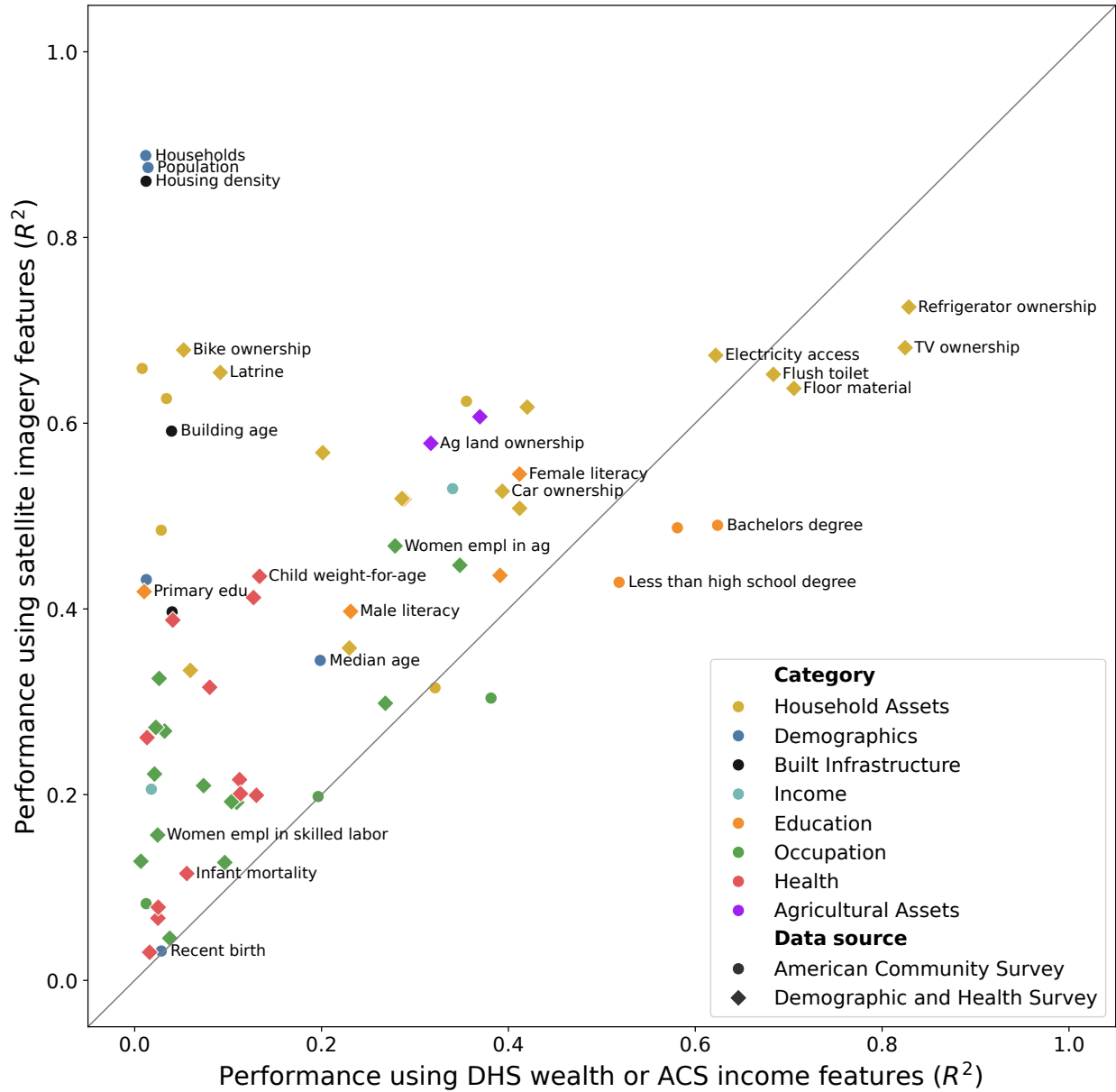


Figure SM6: Comparison of performance between imagery-based models and income or wealth-based models. Figure plots predictive performance (R^2) of imagery-based models from this study (y -axis) against that of models trained using flexible functional forms (quintile bin features) of local income or wealth (x -axis). Each point indicates a single variable for which data are available from either the Demographic and Health Surveys or the American Community Survey. For the former, we use the international wealth index (see D.1) and for the latter we use the log of per capita household income. The 45° degree line is shown in grey. Variable categories are differentiated by color, and data sources are differentiated by symbol. The predominance of points above the 45° line indicates that imagery-based models capture substantially more variation than income (or wealth) alone.

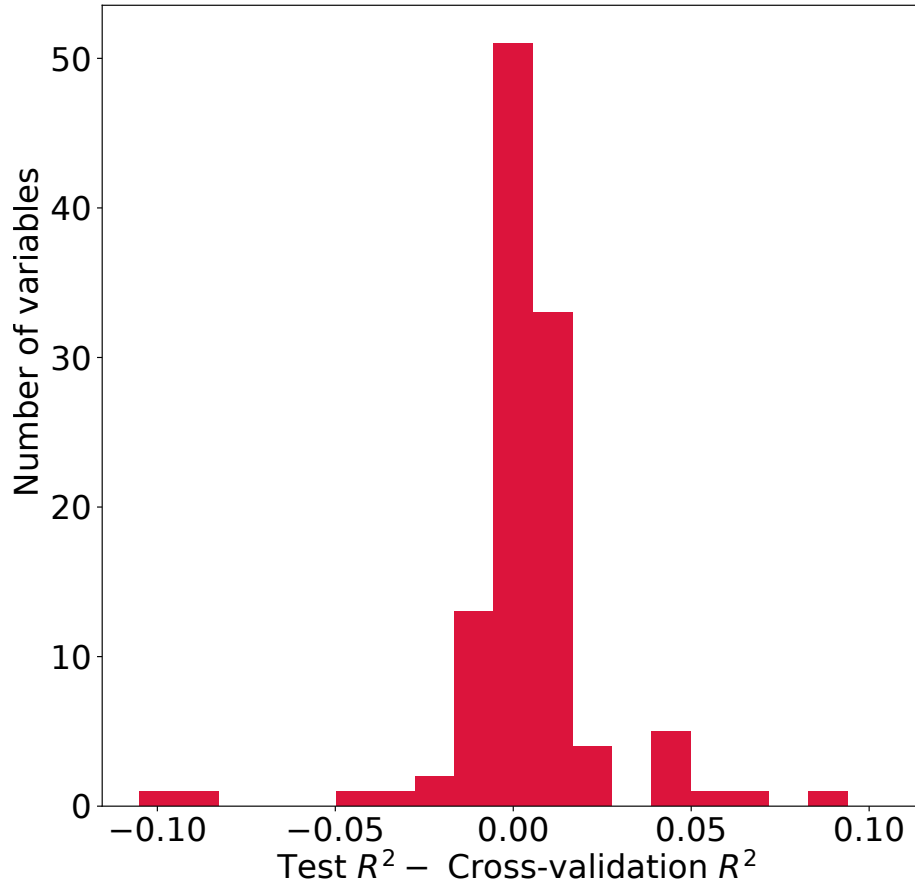


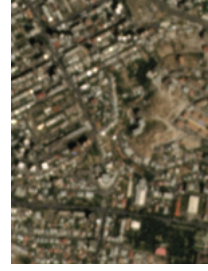
Figure SM7: Distribution of differences in model performance when evaluated on a held-out test set and validation set across 115 variables. Histogram shows the difference in R^2 performance between a completely held-out test set sample of 20% of observations (“Test R^2 ”) and that reported from within the five cross-validation folds used to tune ridge regression penalization parameters in the remaining 80% of the sample data (“cross-validation R^2 ”). Figure shows that differences between these two sets are small and centered on zero. Wherever possible, performance in the held-out test set is reported throughout the main text. In some cases (e.g., scatter plots and maps), performance in the validation set is shown for visual display purposes only.



Figure SM8: Gas and oil pipelines are constructed underground leading to no visual signal in satellite imagery. Figure shows example images of grid cells in the United States with crude oil pipelines (left, pipelines shown in blue), hydrocarbon gas liquids (HGL) pipelines (middle, pipelines shown in orange), and petroleum product pipelines (right, pipelines shown in pink). Because pipelines are built underground, there is no visual signal in satellite imagery that can predict the presence of these pipelines.

C Supplementary Tables

Yerevan, Armenia
(44.505, 40.195)



Varanasi, India
(83.005, 25.395)



Category		Pred- iction	Qual- ity	Pred- iction	Qual- ity
Agricultural Assets	Ag land ownership (% HH)	56	***	57	***
	Livestock ownership (% HH)	1	****	46	****
Agriculture	ln(Global cropped area (%))	-2	****	-0.6	****
	ln(Global irrigated area (%))	-0.9	****	4	****
	ln(Global maize cropped area (%))	-9	****	-5	****
	ln(Global pasture area (%))	-5	****	-11	****
	ln(Global rice cropped area (%))	-11	****	-1	****
	ln(Global soy cropped area (%))	-9	****	-17	****
	ln(Global wheat cropped area (%))	-7	****	-1	****
Built Infrastructure	Road length (km/km2)	19	****	0.9	****
	Expected schooling (yrs)	15	****	11	****
	Female literacy (% pop)	-	-	69	***
	Higher edu (% of pop >= 6 yrs)	75	***	0.8	***
Education	Male literacy (% pop)	-	-	78	*
	Primary edu (% of pop >= 6 yrs)	0.0	*	6	*
	Schooling (yrs)	12	****	6	****
	Secondary edu (% of pop >= 6 yrs)	25	***	4	****
Health	Child height-for-age (z-score)	-0.4	*	-2	*
	Child overweight (% pop <=5yr)	10	*	0.2	*
	Child overweight-for-age (% pop <=5yr)	0.5	*	0.0	*
	Child underweight-for-age (% pop <=5yr)	0.0	**	47	**
	Child weight-for-age (z-score)	0.4	**	-2	**
	Child weight-for-height (z-score)	0.2	**	-1	**
	Life expectancy (yrs)	68	****	66	****
	ln(Child mortality (deaths <=5yr/1000 live births))	3	*	3	*
	ln(Child stunting (% pop <=5yr))	2	*	4	*
	ln(Child wasting (% pop <=5yr))	1	*	3	*
Household Assets	ln(Infant mortality (deaths <=1yr/1000 live births))	3	*	4	*
	ln(Neonatal mortality (deaths <=1mo/1000 live births))	3	*	4	*
	Bike ownership (% HH)	7	****	100	****
	Car ownership (% HH)	62	****	7	****
	Cellphone ownership DHS (% HH)	95	**	86	**
	Electricity access (% HH)	45	**	83	**
	Floor material (index)	1	****	1.0	****
	Flush toilet (% HH)	53	****	0.0	****
	Improved water source (% HH)	100	**	89	**
	Intl Wealth Index (na)	80	****	2	****
Income	Landline telephone (% HH)	100	**	0.0	**
	Latrine (% HH)	0.0	****	45	****
	No sanitation facility (% HH)	64	****	100	****
	Refrigerator ownership (% HH)	100	***	3	***
	Sleeping rooms (per HH)	1	**	2	**
	TV ownership (% HH)	30	****	5	****
	ln(Drinking water access (avg mins))	-2	****	3	****
	Gross Natl Income (\$/pop)	79	****	0.4	****
	Human Dev Index (na)	79	****	44	****
	ln(Nighttime lights radiance (nW/cm2/sr))	6	****	-0.2	****
Natural Systems	Elevation (m)	400	****	-272	****
	Forest cover (%)	3	****	0.0	****
	Gravitational anomaly (milligal)	57	**	-86	**
	Gravitational anomaly change (cm ewh/yr)	0.0	****	0.0	****
Occupation	Ruggedness (m)	0.0	****	0.0	****
	Men empl in ag (%)	0.0	****	12	***
	Men empl in professional (%)	74	*	5	*
	Men empl in service (%)	28	**	22	**
	Men empl in unskilled labor (%)	0.0	**	44	*
	Women empl in ag (%)	6	****	72	***
	Women empl in professional (%)	70	**	0.0	**
	Women empl in service (%)	26	*	18	*
	Women empl in skilled labor (%)	21	*	38	*
	ln(Men empl in clerical (%))	2	*	2	*
	ln(Men empl in skilled labor (%))	2	*	3	*
	ln(Women empl in clerical (%))	1	*	1	*
	ln(Women empl in unskilled labor (%))	1	**	1	*

Table SM1: Predicted variables and data quality scores for two example $0.01^\circ \times 0.01^\circ$ image grid cells. Predicted values and data quality scores for 64 variables in two example locations. Images of each location are shown at the top of the table; the left is located in Yerevan, Armenia and the right in Varanasi, India. Predictions are provided in the units shown in Table 1, with variables modeled in natural logarithm form indicated as $\ln()$. Data quality scores range from 0 (-) to 5 (****), with 5 indicating the highest quality prediction and 0 indicating locations that either are farther than 8° away from ground truth data or are estimated to have no predictive performance (Materials and Methods Section A.6). Only variables with quality scores ≥ 1 (*) are shown.

Variable	Category	Visible (V) or not (N)	Human (H) or not (N)	Seasonal (S) or not (N)	Size (1-3)	Polygon (P) or grid (G)	R^2
Maize yield (kg/ha)	Agriculture	V	H	S	2	P	0.90
Households (HH/km ²)	Demographics	N	H	N	2	P	0.89
Forest cover (%)	Natural Systems	V	N	N	1	G	0.89
Population (pop/km ²)	Demographics	N	H	N	2	P	0.88
Housing density (units/km ²)	Built Infrastructure	V	H	N	2	P	0.86
Soy yield (kg/ha)	Agriculture	V	H	S	2	P	0.80
US nontree cropped area (%)	Agriculture	V	H	S	1	G	0.78
Human Dev Index	Income	N	H	N	3	P	0.78
Building footprint area (%)	Built Infrastructure	V	H	N	1	G	0.78
Tree diversity (species)	Natural Systems	V	N	N	1	G	0.74
Intl Wealth Index	Household Assets	N	H	N	2	P	0.74
Schooling (yrs)	Education	N	H	N	3	P	0.72
Refrigerator ownership (% HH)	Household Assets	N	H	N	2	P	0.72
Life expectancy (yrs)	Health	N	H	N	3	P	0.72
Gross Natl Income (\$/pop)	Income	N	H	N	3	P	0.72
US soy cropped area (%)	Agriculture	V	H	S	1	G	0.70
Global cropped area (%)	Agriculture	V	H	S	1	G	0.70
TV ownership (% HH)	Household Assets	N	H	N	2	P	0.68
Bike ownership (% HH)	Household Assets	V	H	N	2	P	0.68
Electricity access (% HH)	Household Assets	N	H	N	2	P	0.67
Vehicle ownership (per HH)	Household Assets	V	H	N	2	P	0.66
US maize cropped area (%)	Agriculture	V	H	S	1	G	0.66
Latrine (% HH)	Household Assets	V	H	N	2	P	0.66
Road length (km/km ²)	Built Infrastructure	V	H	N	1	G	0.65
Flush toilet (% HH)	Household Assets	V	H	N	2	P	0.65
Expected schooling (yrs)	Education	N	H	N	3	P	0.65
Global pasture area (%)	Agriculture	V	H	S	1	G	0.64
Floor material (index)	Household Assets	N	H	N	2	P	0.64
Homes w LP gas (%)	Household Assets	N	H	N	2	P	0.63
Global rice cropped area (%)	Agriculture	V	H	S	1	G	0.63
No sanitation facility (% HH)	Household Assets	V	H	N	2	P	0.62
Home value (\$)	Household Assets	N	H	N	2	P	0.62
Gravitational anomaly change (cm ewh/yr)	Natural Systems	N	N	N	2	P	0.62
Global maize cropped area (%)	Agriculture	V	H	S	1	G	0.62
Elevation (m)	Natural Systems	V	N	N	1	G	0.62
Ag involvement (% HH)	Agriculture	N	H	N	2	P	0.62
Livestock ownership (% HH)	Agricultural Assets	N	H	N	2	P	0.61
Global wheat cropped area (%)	Agriculture	V	H	S	1	G	0.60
Building age (median yr built)	Built Infrastructure	N	H	N	2	P	0.59
Ag land ownership (% HH)	Agricultural Assets	N	H	N	2	P	0.58
Drinking water access (avg mins)	Household Assets	N	H	N	2	P	0.57
Female literacy (% pop)	Education	N	H	N	2	P	0.55
Bird diversity (species)	Natural Systems	N	N	N	3	P	0.55
Consumption (\$/pop)	Income	N	H	N	2	P	0.54
Ruggedness (m)	Natural Systems	V	N	N	1	G	0.53
Gross rent (\$)	Income	N	H	N	2	P	0.53
Car ownership (% HH)	Household Assets	V	H	N	2	P	0.53
Secondary edu (% of pop ≥ 6 yrs)	Education	N	H	N	2	P	0.52
Nighttime lights radiance (nW/cm2/sr)	Income	V	H	N	1	G	0.52
Landline telephone (% HH)	Household Assets	N	H	N	2	P	0.52
Global irrigated area (%)	Agriculture	V	H	S	1	G	0.52
Global soy cropped area (%)	Agriculture	V	H	S	1	G	0.51
Cellphone ownership DHS (% HH)	Household Assets	N	H	N	2	P	0.51
Income (\$/pop)	Income	N	H	N	2	P	0.49
Graduate degree (% pop ≥ 25 yrs)	Education	N	H	N	2	P	0.49
Food worrying (% HH in past wk)	Health	N	H	N	2	P	0.49
Bachelors degree (% pop ≥ 25 yrs)	Education	N	H	N	2	P	0.49
Bedrooms (per HH)	Household Assets	N	H	N	2	P	0.48
Women empl in ag (%)	Occupation	N	H	N	2	P	0.47
US wheat cropped area (%)	Agriculture	V	H	S	1	G	0.47
Men empl in ag (%)	Occupation	N	H	N	2	P	0.45
Higher edu (% of pop ≥ 6 yrs)	Education	N	H	N	2	P	0.44
Child weight-for-age (z-score)	Health	N	H	N	2	P	0.44
Less than high school degree (% pop ≥ 25 yrs)	Education	N	H	N	2	P	0.43
Food stamps (% HH)	Demographics	N	H	N	2	P	0.43
Food coping (index)	Health	N	H	N	2	P	0.43
Primary edu (% of pop ≥ 6 yrs)	Education	N	H	N	2	P	0.42
Child malaria (%)	Health	N	H	N	2	P	0.41
Male literacy (% pop)	Education	N	H	N	2	P	0.40
Housing units vacant (%)	Built Infrastructure	N	H	N	2	P	0.40
Housing units occupied (%)	Built Infrastructure	N	H	N	2	P	0.40
Child weight-for-height (z-score)	Health	N	H	N	2	P	0.39
Improved water source (% HH)	Household Assets	N	H	N	2	P	0.36
Cellphone ownership LSMS (% HH)	Household Assets	N	H	N	2	P	0.36
Median age (yrs)	Demographics	N	H	N	2	P	0.34
Sleeping rooms (per HH)	Household Assets	N	H	N	2	P	0.33
Gravitational anomaly (milligal)	Natural Systems	N	N	N	1	G	0.33
Women empl in service (%)	Occupation	N	H	N	2	P	0.32
Computer ownership (% pop)	Household Assets	N	H	N	2	P	0.32
Child underweight-for-age (% pop ≤ 5 yr)	Health	N	H	N	2	P	0.32
US cotton cropped area (%)	Agriculture	N	H	S	1	G	0.31
Women empl in professional (%)	Occupation	N	H	N	2	P	0.30
Livestock sales (\$/yr/HH)	Agricultural Assets	N	H	N	2	P	0.30
Empl in service (%)	Occupation	N	H	N	2	P	0.30
Livestock value (\$/HH)	Agricultural Assets	N	H	N	2	P	0.28
Women empl in unskilled labor (%)	Occupation	N	H	N	2	P	0.27
Women empl in domestic (%)	Occupation	N	H	N	2	P	0.27
Child wasting (% pop < 5 yr)	Health	N	H	N	2	P	0.26
Men empl in domestic (%)	Occupation	N	H	N	2	P	0.22
Child mortality (deaths < 5 yr/1000 live births)	Health	N	H	N	2	P	0.22
Women empl in clerical (%)	Occupation	N	H	N	2	P	0.21
Income:poverty ratio	Income	N	H	N	2	P	0.21
Unemployment (%)	Occupation	N	H	N	2	P	0.20
Receiving loan (% HH)	Income	N	H	N	2	P	0.20
Child stunting (% pop ≤ 5 yr)	Health	N	H	N	2	P	0.20
Child height-for-age (z-score)	Health	N	H	N	2	P	0.20
Men empl in skilled labor (%)	Occupation	N	H	N	2	P	0.19
Men empl in service (%)	Occupation	N	H	N	2	P	0.19
Women empl in skilled labor (%)	Occupation	N	H	N	2	P	0.16
US tree cropped area (%)	Agriculture	V	H	S	1	G	0.16
Men empl in unskilled labor (%)	Occupation	N	H	N	2	P	0.13
Men empl in professional (%)	Occupation	N	H	N	2	P	0.13
Infant mortality (deaths ≤ 1 yr/1000 live births)	Health	N	H	N	2	P	0.12
Empl in sales/office (%)	Occupation	N	H	N	2	P	0.08
Child overweight (% pop ≤ 5 yr)	Health	N	H	N	2	P	0.08
Neonatal mortality (deaths ≤ 1 mo/1000 live births)	Health	N	H	N	2	P	0.07
Bee diversity (species)	Natural Systems	N	N	N	1	G	0.06
Rejected for loan (% HH)	Income	N	H	N	2	P	0.04
Men empl in clerical (%)	Occupation	N	H	N	2	P	0.04
Recent birth (% women)	Demographics	N	H	N	2	P	0.03
Child overweight-for-age (% pop ≤ 5 yr)	Health	N	H	N	2	P	0.03
Produce sales (\$/yr/HH)	Agricultural Assets	N	H	N	2	P	0.01
Petroleum pipelines (km/km ²)	Built Infrastructure	V	H	N	1	G	0.00
HGL pipelines (km/km ²)	Built Infrastructure	V	H	N	1	G	0.00
Crude oil pipelines (km/km ²)	Built Infrastructure	V	H	N	1	G	0.00

Table SM2: Variable categories, properties and predictive performance. Following Figure 2, each variable is categorized: into one of 10 categories; as directly visible or not; as human or non-human; as seasonal or not; and into three groups listing the average size of the ground truth observations (1: <1km², 2: 2-499km², or 3: ≥500km²). Variables are modeled either as polygons or grids (Materials and Methods Section A.2) and are ordered here by performance.

Variable	Performance in this study (R^2)	Performance in prior literature (as reported therein ¹)	Spatial coverage of prior prediction	Prior model includes additional covariates?	Reference
Maize yield	0.90	Adjust $r^2 = 0.39^2$	1 country	No	(56)
Forest cover	0.89	$R^2 = 0.85$	Global	No	(7)
Households	0.89	$R^2 = 0.88-0.96^3$	2 countries	No	(57)
Population	0.88	$r^2 = 0.9365^3$	1 country	No	(58)
Housing density	0.86	$R^2 = 0.22$	1 country	No	(7)
Soy yield	0.80	RMSE = 0.49-1.04	2 countries	No	(59)
Building footprint area	0.78	Correctness at breakeven = 94.6 ⁴	Subnational	No	(60)
US nontree cropped area	0.78	Overall accuracy = 0.8554 ⁵	Subnational	No	(61)
Intl Wealth Index	0.74	Pearson $r^2 = 0.63$	48 countries	No	(13)
Tree diversity	0.74	"Explained variation" = 88%	Global	Yes	(47)
Schooling	0.72	Pearson $r^2 = 0.26$	53 countries	No	(13)
Gross Natl Income	0.72	Adj. R-squared = 0.2492-1.00 ³ , 0.8971 ⁶	26 countries	Yes	(62)
Global cropped area	0.70	Overall accuracy = 99.1	Global	No	(63)
US soy cropped area	0.70	R ² =0.945	Subnational	No	(64)
Bike ownership	0.68	$R^2 = 0.48$	Subnational	No	(12)
Electricity access	0.67	$R^2 = 0.24 - 0.69^{3,7}$	4 countries	No	(11)
US maize cropped area	0.66	Exceeds 85% accuracy for 2008-2018 ⁸	Subnational	No	(59)
Latrine	0.66	Pearson $r^2 = 0.36$	49 countries	No	(13)
Flush toilet	0.65	Pearson $r^2 = 0.36$	49 countries	No	(13)

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Variable	Performance in this study (R^2)	Performance in prior literature (as reported therein ¹)	Spatial coverage of prior prediction	Prior model includes additional covariates?	Reference
Road length	0.65	$R^2 = 0.52$	1 country	No	(7)
Global pasture area	0.64	Producer's accuracy = 94.6%; user's accuracy = 91.8% ⁹	Subnational	No	(65)
Global rice cropped area	0.63	80% (classification of all rice classes), $R^2 = 97\%$	6 countries	Yes	(66)
Homes w LP gas	0.63	$R^2 = 0.57$	Subnational	No	(12)
Global maize cropped area	0.62	R2=0.945	Subnational	No	(64)
Elevation	0.62	$R^2 = 0.45$	Global	No	(7)
Home value	0.62	$R^2 = 0.48$	1 country	No	(7)
No sanitation facility	0.61	Pearson $r^2 = 0.36$	49 countries	No	(13)
Global wheat cropped area	0.60	Producer's accuracy: 95.3, user's accuracy: 96.1	Subnational	No	(65)
Building age	0.59	$R^2 = 0.52$	1 country	No	(7)
Drinking water access	0.57	$R^2 = 0.11-0.34^{3,7}$	4 countries	No	(11)
Female literacy	0.55	$R^2 = 0.7$	Subnational	No	(12)
Bird diversity	0.55	$R^2_{adj} = 0.51$	1 country	No	(67)
Car ownership	0.53	$R^2 = 0.65$	Subnational	No	(12)
Global irrigated area	0.52	Overall accuracy = 0.96	Global	Yes	(68)
Nighttime lights radiance	0.52	$R^2 = 0.49$	Global	No	(7)
Global soy cropped area	0.51	Producer's accuracy = 86.9%; user's accuracy = 89.1%	Subnational	No	(65)

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Variable	Performance in this study (R^2)	Performance in prior literature (as reported therein ¹)	Spatial coverage of prior prediction	Prior model includes additional covariates?	Reference
Cellphone ownership DHS	0.51	Pearson $R^2 = 0.151$ - 0.169 ⁶	2 countries	No	(69)
Secondary edu	0.51	$R^2 = 0.47 - 0.64$ ³	4 countries	No	(11)
Income	0.49	$R^2 = 0.45$	1 country	No	(7)
Consumption	0.49	$r^2 = 0.44$ ⁶	5 countries	No	(3)
Bachelors degree	0.49	$R^2 = 0.38$	1 country	No	(7)
US wheat cropped area	0.47	Producer's accuracy = 95.4%; user's accuracy = 96.1% ⁹	Subnational	No	(65)
Higher edu	0.44	$R^2 = 0.47 - 0.64$ ³	4 countries	No	(11)
Child weight-for-age	0.44	$R^2 = 0.01 - 0.36$ ^{3,7}	4 countries	No	(11)
Food coping	0.43	$R^2 = 0.119, 0.337$ ¹⁰	1 country	Yes	(70)
Food stamps	0.43	$R^2 = 0.3$	1 country	No	(7)
Primary edu	0.42	$R^2 = 0.47 - 0.64$ ³	4 countries	No	(11)
Child malaria	0.41	For Africa: mean error = -0.9; mean absolute error = 11.98; correlation = 0.86 ¹¹	Global	Yes	(71)
Housing units vacant	0.40	$R^2 = 0.26$	1 country	No	(7)
Male literacy	0.40	$R^2 = 0.35$	Subnational	No	(12)
Child weight-for-height	0.39	$R^2 = -0.02 - 0.11$ ^{3,7}	4 countries	No	(11)
Improved water source	0.36	Pearson $r^2 = 0.40$	49 countries	No	(13)
Cellphone ownership LSMS	0.36	Pearson $R^2 = 0.268$ - 0.414 ³	2 countries	No	(69)
Median age	0.34	$R^2 = 0.87$ - 0.96 ³	2 countries	No	(57)

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Variable	Performance in this study (R^2)	Performance in prior literature (as reported therein ¹)	Spatial coverage of prior prediction	Prior model includes additional covariates?	Reference
US cotton cropped area	0.31	Overall accuracy = 92.5, Kappa = 0.881	Subnational	No	(72)
Child underweight-for-age	0.31	$R^2 = 0.28$	Subnational	No	(12)
Child wasting	0.26	$R^2 = 0.04$	Subnational	No	(12)
Child mortality	0.22	Pearson $r^2 = 0.01$	56 countries	No	(13)
Unemployment	0.20	Accuracy: 4.9%-8.0%	1 country	Yes	(73)
Child height-for-age	0.20	$R^2 = 0.02 - 0.36^{3,7}$	4 countries	No	(11)
US tree cropped area	0.16	Overall accuracy = 0.736 ¹²	Subnational	No	(74)
Infant mortality	0.12	Mean error = -0.0018; mean absolute error = 0.0183; correlation = 0.604; 95% coverage = 96.8% ¹¹	LMIC ¹³	Yes	(53)
Child overweight	0.08	$r^2 = 0.17^6$	11 countries	Yes	(75)
Neonatal mortality	0.07	Mean error = -0.000866; mean absolute error = 0.0125; correlation = 0.414; 95% coverage = 96.6% ¹¹	LMIC ¹³	Yes	(53)
Bee diversity	0.06	$r^2_{adj} = 0.592$	Global	Yes	(54)
Recent birth	0.03	Not reported	1 country	No	(76)

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Variable	Performance in this study (R^2)	Performance in prior literature (as reported therein ¹)	Spatial coverage of prior prediction	Prior literature includes additional covariates?	Reference
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¹ Performance metrics are reported using the terms provided in the original reference. Though we do our best to harmonize across studies; performance metrics reported in other studies may not be directly comparable to the R^2 performance we report for the tasks analyzed in this study (Materials and Methods Section A.5).

² Uses calculated indices as inputs.

³ Country-specific model (i.e., model was trained and evaluated using data from the same country for which it generated predictions).

⁴ For proposed model at highest level of geographic aggregation.

⁵ This paper looks at several nontree crops, but not tree crops as a category itself.

⁶ Pooled model (i.e., model was trained using data from multiple countries).

⁷ Performance estimated based on graphic in original reference.

⁸ Model masks non-cropland areas.

⁹ This is one class out of many, performance is for specific class.

¹⁰ Using 2010 data to predict 2013 and 2010, respectively.

¹¹ We assume “correlation” refers to the square root of Pearson’s r^2 . Coverage is the percentage of empirical observations that fell between the 2.5 and 97.5% simulated prediction.

¹² Only for olive orchards.

¹³ Low- and middle-income countries.

Table SM3: Comparison of performance in this study to performance in prior research predicting similar variables using satellite imagery. Table lists all variables for which a similar variable has been predicted from satellite imagery in prior research at the scale of a single country or larger. Performance in this study is reported as out-of-sample R^2 , while various alternative metrics are used in the literature (Materials and Methods Section A.5). Only variables whose performance was previously evaluated using r^2 or R^2 are shown in Figure SM1. If a prior study reports multiple performance values, Figure SM1 shows the average of those values. Variables are listed in order of performance in this study.

D Supplementary Text: Data sources appendix

D.1 (30) Demographic and Health Surveys

We use household data collected via the United States Agency for International Development’s Demographic and Health Surveys (DHS) program for 48 variables. The DHS program has been active since 1984 and has collected nationally-representative data from over 90 countries on topics such as mortality, gender, nutrition, wealth, and sustainable development. DHS data include information collected via interviews based on standard model questionnaires, as well as biomarker collection (e.g., for height, weight, malaria prevalence). We derive labels from three types of questionnaires: Demographic and Health Surveys, AIDS Indicator Surveys, and Malaria Indicator Surveys (30).

DHS surveys are year- and country-specific, and the full suite of DHS variables are not consistently available across all years and countries. Thus, each DHS variable is associated with its own set of surveys. For each variable, we use survey data from the latest set of surveys available during the time of data collection (September, 2022) that include the variable, using only those surveys completed by 2012 or later. In addition, we only include countries for which DHS has granted permission for our team to geolocate the coordinates of survey enumeration areas.

For each survey, DHS provides microdata at the household level via data tables available for download at <https://dhsprogram.com/data/>. Within these tables, DHS associates each household with an enumeration area called a cluster. To protect privacy, DHS randomly displaces the true coordinates of the cluster by up to 2km for urban clusters and up to 5km for rural clusters, with a random 1 percent of rural cluster coordinates displaced by up to 10km. The displaced coordinates fall within the same administrative boundaries as the actual location.

Because the exact locations of each cluster are displaced, we assign each household’s data to a cluster polygon. We create these cluster polygons by generating a buffer of the appropriate radius (2km for urban clusters and 10km for rural clusters) around each cluster centroid. We clip the boundaries of each polygon to the administrative boundary associated with the cluster. Administrative boundaries are provided by the DHS; in some cases they correspond to province or district boundaries, while in others, they represent other geographic aggregations. Like the surveys themselves, boundary files are country- and year-specific and have been obtained from <https://spatialdata.dhsprogram.com/boundaries/> (77). If multiple sets of boundaries are associated with a survey, we use the most finely-resolved boundaries. If boundaries are not available for a given survey, we use the most recent available DHS boundaries associated with a different survey for the same country.

To provide consistency with DHS program metrics, unless otherwise noted, we calculate DHS variables using the equations and variables specified in the *Guide to DHS Statistics* (78). As a validation step, we use DHS-provided cluster weights to calculate outcomes at the national level, which we compare to corresponding national statistics reported on the DHS Program STATcompiler (<https://www.statcompiler.com/>).² The processes for calculating each individual outcome variable are described below.

- *Electricity access; bike ownership; car ownership; landline telephone; cellphone ownership; refrigerator ownership; TV ownership; agricultural land ownership; livestock ownership.* For all variables except electricity access, we calculate the percentage of households in each DHS cluster that own the given asset or durable product. For the electricity access, we calculate the percentage of households within each DHS cluster that have access to electricity in their homes.
- *Floor material.* We calculate average household floor quality using a similar approach to that which ref. (79) uses to calculate the International Wealth Index (IWI). Specifically, we assign each household a floor quality score on a scale of 1-3, with 1 representing “low quality” floor material (none, earth, dung, and other natural materials), 2 representing middle quality floor materials (rudimentary materials, such as cement, concrete, wood, bamboo etc.), and 3 representing high quality floor materials (finished floor with parquet, carpet, tiles, linoleum, ceramic etc.). This variable is defined as the average score across all households surveyed in each DHS cluster.
- *Flush toilet; latrine; no sanitation facility.* We create three variables to indicate the prevalence of different types of household sanitation facilities in each cluster. *Flush toilet* is calculated as the percentage of households using any type of flush toilet (e.g., onsite or offsite flush to piped system, septic tank, or open pit). *Latrine* is calculated as the percentage of households using any type of pit toilet latrine (e.g., pit latrine with or without slab). *No sanitation facility* is calculated as the percentage of households with no toilet facility (e.g., outdoors, bush).
- *Sleeping rooms.* We calculate the average number of rooms for sleeping per household by adopting an approach similar to that used to compute IWI in ref. (79). The number of rooms for sleeping are classified on a scale of 1-3, where 1 indicates 0 or 1 room, 2 represents 2 rooms, and 3 represents 3 or more rooms. The outcome is the average score across all households in each DHS cluster.

²We do not apply weights to the cluster-level metrics that we use as outcome variables.

- *Improved water source.* This variable is calculated as the percent of households in each cluster using an improved source of drinking water. Improved and unimproved water sources reflect the categorizations suggested by the DHS program (78). Improved water sources include piped, bottled, public tap/standpipe, protected well, tanker truck, cart with small tank, and rainwater. Unimproved water sources include unprotected well, river/dam/lake/ponds/stream/canal/irrigation channel, unprotected spring, and other.
- *Drinking water access.* This variable is the average reported time (in minutes) for members of a household to obtain drinking water, where averages are computed across each cluster. We assign households with water on the premises or delivered to the dwelling a time of zero minutes.
- *Child height-for-age; child weight-for-age; child weight-for-height; child overweight; child overweight-for-age; child stunting; child underweight-for-age; child wasting.* We create these eight variables using the anthropometric Z -scores reported in DHS surveys for children under five years of age. These metrics are indicators of child nutritional status. Individual Z -scores for height-for-age, weight-for-height, and weight-for-age are calculated by the DHS program based on World Health Organization (WHO) Child Growth Standards using data on sex, age, height, and weight, as described in (78). For height-for-age, weight-for-age, and weight-for-height, we compute the mean Z -score in each cluster for these individual-level outcomes. For stunting, underweight-for-age, and wasting, we calculate the percentage of children in each cluster whose Z -scores render them moderately or severely stunted, underweight-for-age, or wasted, using a cutoff of Z -scores that fall below minus two (-2.0) standard deviations below the mean height-for-age (stunting), weight-for-age, (underweight-for-age) and weight-for-height (wasted), respectively. The overweight and overweight-for-age variables represent the percentage of children in each cluster who are overweight or overweight for their age, computed as the percentage of Z -scores that fall above two ($+2.0$) standard deviations above the mean weight-for-height and weight-for-age, respectively.
- *Child mortality; infant mortality; neonatal mortality.* We generate three variables describing different metrics of early childhood mortality. In each case, we calculate mortality using the synthetic cohort life table methodology described in ref. (78). Mortality is expressed as a rate per 1,000 live births, with “child” referring to children less than 5 years of age,³ “infant” referring to infants less than or equal to 1 year of age,

³In the DHS data, this category is referred to as “under-five mortality.”

and “neonatal” referring to infants less than or equal to 1 month of age. We calculate these mortality rates by household based on births over the ten-year period preceding the survey date.

- *Child malaria.* This variable is calculated as the percentage of children of age 6-59 months who are classified as having malaria according to a rapid diagnostic test (RDT). Cluster-level outcomes are calculated by dividing the number of children in the cluster testing positive for malaria using an RDT by the total number of children in the cluster tested using an RDT.
- *Higher education; primary education; secondary education.* We generate three variables that describe educational attainment of household members over the age of 6 in the DHS. Primary education is calculated as the percentage of each cluster’s population of age 6 years or older whose highest educational attainment is primary school. Secondary education is calculated as the percentage of each cluster’s population of age 6 years or older whose highest educational attainment is secondary school. Higher education is calculated as the percentage of each cluster’s population of age 6 years or older whose highest educational attainment is beyond secondary school.
- *Male literacy; female literacy.* We calculate one variable each for the percentage of women and men in each cluster who are literate, as reported in the DHS’s Individual Woman’s or Man’s Questionnaire. The DHS program defines literacy as being able to read a part or the entirety of a sentence, or having attained a level of schooling higher than secondary education.
- *Men employed in ag; men employed in clerical; men employed in domestic; men employed in professional; men employed in service; men employed in skilled labor; men employed in unskilled labor; women employed in ag; women employed in clerical; women employed in domestic; women employed in professional; women employed in service; women employed in skilled labor; women employed in unskilled labor.* We generate these 14 occupation variables using the DHS Individual Woman’s or Man’s Questionnaires. Each variable is calculated as the percentage of women or men (in most cases, age 15-49 years) who were employed in the 12 months preceding the survey in the following occupations: professional, technical, or managerial (“professional”); clerical services (“clerical”); sales and services (“service”); skilled manual labor (“skilled labor”); unskilled manual labor (“unskilled labor”); domestic services (“domestic”); or agriculture (“ag”).

- *International Wealth Index (IWI)* IWI is an asset-based wealth index designed to provide a comparable measure of wealth across different countries (79). The index is expressed as a value between 0 and 100, with higher values indicating a higher level of wealth. This index is computed from DHS household data on ownership of consumer durables, housing characteristics, and access to basic services (e.g., water and electricity). The Global Data Lab (GDL) provided us these processed IWI values at the household level, with permission from the DHS program. The cluster value is the mean IWI for the households in each cluster.

D.2 (37) Building footprints

We use data on building footprints provided by Microsoft (37), which supplies over 1.25 million building footprint polygon geometries in GeoJSON vector format for Canada, Tanzania, Uganda, and the United States. Microsoft used deep neural networks to detect building footprints from aerial imagery available to Bing Maps. After removing noise and suspicious data, they applied a polygonization algorithm to detect building edges and angles and creates polygons. We use this data source to construct one variable, described below.

- *Building footprint area.* We use building footprint polygons to calculate the percentage of each $0.01^\circ \times 0.01^\circ$ grid cell that is covered by buildings.

D.3 (32) American Community Survey (ACS)

We use household survey data collected and published by the United States Census Bureau. Specifically, we use 2019 American Community Survey (ACS) 5-year estimates (32). Five-year estimates are considered to have better statistical reliability for sparsely populated areas. All ACS data are provided and analyzed at the census tract level. There are $\approx 74,000$ census tracts in the United States with an average size of $\approx 130 \text{ km}^2$. Census tracts are designed to contain roughly the same population, implying that their geographic size is dependent on population density. More information on the calculation of each individual variable is given below:

- *Building age; Home value; Income; Income:poverty ratio; Median age; Gross rent.* These variables are taken directly from ACS tables with no additional processing. Each of these variables either is or is based on a median value within a census tract.
- *Households; Housing density; Population.* Each of these variables is based on ACS counts within a census tract. In order to convert them to an intensive margin, we divide by the area (km^2) of the respective census tract (e.g., population / km^2).

- *Housing units occupied; Housing units vacant.* These two housing unit count variables are both divided by the number of total housing units to create a percent variable.
- *Food stamps.* ACS provides the total number of households receiving food stamps in the past 12 months. This is divided by the number of households to create a percent indicator.
- *Bachelors degree; Graduate degree; Less than high school degree.* ACS provides these totals as headcounts. To convert to a percent variable, we divide by the total population 25 years of age and older for each census tract.
- *Recent birth.* ACS provides the total number of women who have had a birth in the past 12 months. To convert this to a percent indicator, we divide by the total number of women between the age of 15 and 50 in each census tract.
- *Bedrooms.* The average number of bedrooms per occupied housing unit is calculated as the sum of bedrooms from owner and renter occupied housing units. Housing units with more than 5 bedrooms only get a value of 5 due to data collection limitations. This sum is then divided by the number of occupied housing units.
- *Computer ownership.* To calculate computer ownership in percentage terms, we divide the population owning a computer (across all age groups) by the total population.
- *Homes w LP gas.* This indicator is for occupied housing units with bottled, tank, or liquid petroleum gas. To convert this to a percent variable, we divide the count by the number of occupied units.
- *Vehicle ownership.* This variable is the sum of all vehicles across households divided by the number of households. Vehicle ownership is capped at 4 per household in the original survey data.
- *Unemployment rate.* The unemployment rate is calculated as the total number of men and women who are reported as unemployed divided by the total population over 16 years of age.
- *Empl in sales/office; Empl in service.* These two variables give a count for the number of workers employed in these respective job areas. To create these as percent variables, they are divided by the total numbers of workers 16 and over in each census tract.

D.4 (33) Crop and Pasture Data

We use two datasets provided by Earthstat (33). Both are available at a 5 arcminute (0.08°) resolution and derived from combining agricultural inventory data with satellite imagery.

- *Global cropped area; Global pasture area.* These two labels come from the EarthStat data product called Cropland and Pasture Area in 2000 (80). This data product provides data on area covered by any crop or pasture on all continents except Antarctica.
- *Global maize cropped area; Global rice cropped area; Global soy cropped area; Global wheat cropped area.* These 4 crop-specific variables come from the EarthStat data product called Harvested Area and Yield for 4 Crops (1995-2005). This data product covers 51 countries across the globe and we use only the latest year of data in this analysis.

D.5 (81) FAO Global Information System on Water and Agriculture

For global irrigation data we use data from FAO’s Global Information System on Water and Agriculture (34). This dataset provides percent area equipped for irrigation, available at a 5 arcminute (0.08°) resolution.

To construct this dataset, FAO first compiled administrative data on irrigation using national and subnational survey statistics in combination with irrigation statistics available in reports from FAO, World Bank, and other international organizations. In order to spatially distribute irrigated area within subnational units, FAO relied on digitized irrigation maps from various sources. They also incorporated other digital maps and inventories constructed via remote sensing. A complete description of the FAO methodology was released in conjunction with the 3rd version of their data product (81).

- *Global irrigated area.* Irrigation data for the year closest to 2005 were used by FAO when constructing this variable and the data product should be understood to represent the spatial extent of area equipped for irrigation in the year 2005.

D.6 (42) North American Breeding Birds Survey

We use data from the North American Breeding Bird Survey (BBS) conducted in 2019 (42).⁴ The survey comprises avian point counts along randomly established roadside survey

⁴We also acknowledge the thousands of dedicated U.S. and Canadian participants who annually perform and coordinate the survey.

routes in all U.S. states except Hawaii and all Canadian provinces except Nunavut. Each survey route is approximately 39.2km long, and BBS provides the latitude and longitude coordinates of each route’s starting point. At 800-meter intervals along each route, a skilled citizen scientist records every bird seen or heard within 400m for a three-minute period. Each route was surveyed once in 2019 during breeding season, with survey months ranging from April through July. We use data only from the 2,891 routes that meet the Breeding Bird Survey’s data quality criteria, namely, that the data collection adhered to the BBS sampling protocol; that conditions meet BBS-defined criteria for weather, data, time, and route completion; and that the survey took place on a randomly established route. We use this data to construct one variable:

- *Bird diversity.* Bird diversity expresses breeding bird species richness, i.e., the total number of unique taxa (as indicated in the BBS data by the 5-digit “AOU” code originally derived from American Ornithologists’ Union species identification codes) observed along each route. Since only the starting points and not the trajectories of each route are known, we use the average length of survey routes (39.2km) to define a buffer representing the plausible area for bird sightings around each starting point. We analyze species richness using these polygon buffers as the unit of analysis.

D.7 (39) Open Street Maps

In order to calculate road length at the global level, we queried the Open Streets Maps (OSM) API (39). OSM provides crowdsourced data on roads, routes, and paths that connect geographic areas (82). According to ref. (83), OSM is the only global, openly-licensed source of road data and it is estimated to be approximately 83% complete (83).

- *Road length.* To construct our global dataset of roads, we first identified the subset of OSM “paths” that are broadly consistent with the general understanding of a “road.” For example, this excludes pedestrian footpaths, but includes runaway truck ramps. We then individually queried every grid cell from the global 1% sample of our 0.1° resolution grid using the OSM API. We then calculated the length of road in each grid cell, and divided by the total area of the grid cell.

D.8 (36) Cropland Data Layer

For labels related to US agricultural land cover, we use the United States Department of Agriculture (USDA) Cropland Data Layer (CDL) for the year 2019 (36). The CDL provides agricultural land cover information for the continental United States at a 30m resolution. The

CDL is produced on annual basis through a combination of remote sensing and agricultural ground truth data collected during the growing season.

This data is at a finer resolution than our $0.01^\circ \times 0.01^\circ$ grid. For each grid cell, we count the number of 30m x 30m pixels that correspond to each crop and divide by the total number of source pixels within the given grid cell. Raster pixels are not divided into partial pixels when they partially overlay the grid cell. Instead we add a small amount of outward buffering, meaning that the true area of consideration is slightly larger than the grid cell. Along the coast of the continental US, there are some alignment issues between the raster data and the $0.01^\circ \times 0.01^\circ$ grid. We resolve this issue by dropping these grid cells.

More information on specific labels is given below:

- *US cotton cropped area; US maize cropped area; US soy cropped area; US rice cropped area; US wheat cropped area.* Each of these crop types is represented by 1-3 specific numerical values in the CDL raster product.
- *US nontree cropped area; US tree cropped area.* Tree and nontree crops are represented by a range of values in the CDL raster. For these labels we consider any crop type that falls into these broad categories of crop types, defined in the CDL reference material.

D.9 (79) Global Data Lab (GDL)

The Global Data Lab provides human development indicators at the first administrative level (e.g., U.S. states). These large administrative polygon labels vary greatly in shape and extent (19). GDL derives these statistics from a combination of survey sources including Eurostat, DHS, Multiple Indicator Cluster Surveys (MICS), and Integrated Public Use Microdata Series (IPUMS) (31). GDL also ensures that population weighted averages are centered to match the official United Nations Development Program indicators that are released each year.

- *Expected Schooling; Gross Natl Income; Human Dev Index; Life Expectancy; Schooling.* The Human Development Index (HDI) is a measure of human wellbeing that captures not only income but also education and health, and is thus commonly used as an alternative to economic growth (84). The GDL provides data on HDI as well as on the indices used to create HDI: gross national income per capita, life expectancy at birth, expected years of schooling, and mean schooling. We use data for the year 2018, the most recent data available when this project began. We note that the GDL is not able to measure each HDI component directly at the subnational level. Instead,

they sometimes have to estimate subnational variation. For example, all of the within-country variation in GNI is estimated based on the within-country variation of the International Wealth Index (IWI).

D.10 (41) Native Bee Inventory and Monitoring Lab

We use data from the US Geological Survey Eastern Ecological Science Center Native Bee Laboratory (41), which provides species occurrence records for native and non-native bees, wasps, and other insects that were collected using a variety of trapping methods. Each record includes information about the sample (i.e., insect), its taxonomic identification, and the latitude and longitude at which it was collected. The database includes only records of species presence (absences are not recorded). We construct the following variable:

- *Bee diversity.* We use the point data to compute a count of the unique species documented by all sampling events between 2009 and 2019 that occurred within each $0.01^\circ \times 0.01^\circ$ grid cell in North America (including U.S. territories and Minor Outlying islands). If more than one sampling event took place within a grid cell, the variable represents the count of unique species across events (i.e., species are not double-counted).

D.11 (38) Energy Information Association

We create three variables using data from the Energy Information Association (38).

- *Crude oil pipelines, HGL pipelines, Petroleum pipelines.* These variables measure the length in kilometers of three types of pipelines in a grid cell: crude oil pipelines, hydrocarbon gas liquids (HGL) pipelines, and petroleum product pipelines. The data include interstate trunk lines and selected intrastate lines as provided by the Energy Information Association (EIA). Data are from January 2020 and cover the lower 48 United States. As shown in Figure SM8, these pipelines are constructed underground, leading to no visual signal observable from satellite imagery.

D.12 (40) Nighttime Lights

We use nighttime luminosity (average radiance) data provided by the Earth Observations Group at the National Oceanic and Atmospheric Administration and the National Geophysical Data Center. This data is derived from satellite images captured at night by the Visible Infrared Imaging Radiometer Suite (VIIRS), with radiance units of nanowatts per square centimeter per steradian ($\text{nWcm}^{-2}\text{sr}^{-1}$). The raw raster data covers the globe from a latitude

of 75N to 65S, and we use annual composite values from the year 2015 to construct a single variable as described below.

- *Nighttime lights*. The ground truth data has a resolution of 15 arc-seconds (0.004°). To construct this variable, we perform a spatial aggregation of the raster data by taking the average of all ground truth values within each $0.01^\circ \times 0.01^\circ$ grid cell on the sparse sampled grid. The resulting value is mean nighttime lights radiance for each sparse grid cell.

D.13 (2) Forest Cover

We use a raster dataset from ref. (2) that measures global forest cover. The authors define forest as all vegetation taller than 5m in height. To derive these estimates of annual forest cover, a model was used based on Landsat imagery captured during the growing season. We use values from 2010 to construct the percent of each grid cell covered with forest:

- *Forest Cover*. The ground truth data has a finer resolution ($30\text{m} \times 30\text{m}$) than the sparse sampled grid. To construct this variable, we aggregate the raster data by taking the average percent of forest cover within each $0.01^\circ \times 0.01^\circ$ grid cell on the sparse sampled grid.

D.14 (46) Ruggedness

We use a raster dataset developed by ref. (46) comprising global, geo-referenced data on terrain ruggedness. To construct this dataset, the authors divided all terrestrial areas of the world into identical grid cells with a resolution of 30 arc-seconds (approximately $1\text{km} \times 1\text{km}$). Next, using absolute elevation data from the Land Processes Distributed Active Archive Center, they measured absolute elevation change between each grid cell and all contiguous grid cells. Hence, the ruggedness of any given grid cell is determined by measuring how the average elevation of that grid cell differs from all neighboring grid cells. Each such difference was squared so that positive and negative elevation changes contributed equally to the overall ruggedness measure. Using this dataset, we construct a single variable as described below.

- *Ruggedness*. To create this variable, we spatially overlay the global 1% sampled grid on the raw ruggedness raster dataset. We then perform an aggregation of the raster data by taking the average of all ground truth values that overlap with each $0.01^\circ \times 0.01^\circ$ grid cell. The resulting value is mean ruggedness within each sparse grid cell.

D.15 (43) Elevation

We obtain elevation data from Mapzen via Amazon Web Services (AWS) Open Data Terrain Tiles. This is a global dataset that provides basemap terrain coverage in raster tile format. The underlying data sources for Mapzen Terrain Tiles vary, comprising:

- ArcticDEM terrain data DEM(s), created from DigitalGlobe, Inc., imagery and funded under National Science Foundation awards 1043681, 1559691, and 1542736;
- Australia terrain data © Commonwealth of Australia (Geoscience Australia) 2017;
- Austria terrain data © offene Daten Österreichs – Digitales Geländemodell (DGM) Österreich;
- Canada terrain data, containing information licensed under the Canadian Open Government Licence;
- Europe terrain data produced using Copernicus data and information funded by the European Union - EU-DEM layers;
- Global ETOPO1 terrain data, U.S. National Oceanic and Atmospheric Administration
- Mexico terrain data, INEGI, Continental relief, 2016;
- New Zealand terrain data; Copyright 2011 Crown copyright (c) Land Information New Zealand and the New Zealand Government (All rights reserved);
- Norway terrain data © Kartverket;
- United Kingdom terrain data © Environment Agency copyright and/or database right 2015. All rights reserved;
- United States 3DEP (formerly NED) and global GMTED2010 and SRTM terrain data courtesy of the U.S. Geological Survey.

We use this data source to construct a single variable.

- *Elevation.* This data can be accessed through AWS at different zoom levels, ranging from 1 to 14. We use zoom level 8 (a ground resolution of approximately 611.5m at the equator). We directly access this data using the `get_aws.terrain()` function via the R package `elevatr`. As the ground truth data has a finer resolution than 0.01° , we perform an aggregation of the raster data by taking the average of all ground truth values that fall in each $0.01^\circ \times 0.01^\circ$ grid cell on the 1% sampled sparse grid. The resulting value is mean elevation within each sparse grid cell.

D.16 (47) Tree diversity

We use data on tree species richness that have been derived from a variety of sources and presented together in ref. (47). While the authors include species richness for different spatial extents, we use only information from “plots” in our analysis, which are approximately the size of a single $\approx 1 \text{ km}^2$ grid cell.

- *Tree diversity.* We use the number of unique tree species identified for various plots and match them to their nearest grid cell centroid.

D.17 (35) Crop yields

We obtain administrative maize and soybean yield data from the United States, China, Brazil and the European Union from ref. (35). Most of the data is at the second administrative level (e.g., U.S. counties).

- *Maize yield, Soy yield.* Yields are calculated as harvested grain divided by the planted area. Sometimes harvested area is used in place of planted area. See ref. (35) for details.

D.18 (45) GRACE Trends

GRACE trends data were download from NASA but developed and presented in refs. (45, 85).

- *Gravitational anomaly change.* This variable indicates the average change per year in the observed gravitational anomaly from the year 2002 to 2016. Changes are calculated for a 1° equal area grid. This grid is far coarser than the 0.01° grid used here, so we treat each grid cell as a polygon level observation.

D.19 (44) GRACE Levels

Global gravity level data were downloaded from the International Centre for Global Earth Models (ICGEM) (44). We use gravity levels from the GIF48 model which is discussed in ref. (86). Generally, this gravity model uses observed but coarse gravity readings from the GRACE satellite and combines it with more fine resolution elevation and terrestrial data in order to estimate a gravity anomaly field at fine resolution (0.1°).

- *Gravitational anomaly.* We use the estimated gravity anomaly from the GIF48 model. We model this gravity anomaly field on the global 1% sample sparse grid.

D.20 (29) Living Standards Measurement Surveys

Living Standards Measurement Surveys - Integrated Surveys on Agriculture (LSMS) are conducted at the country level and their data are made available by the World Bank via worldbank.org/en/programs/lsms (29). The variables used here come from the following set of country-level LSMS surveys: Ethiopia (2018), Malawi (2016), Niger (2011), Nigeria (2015 & 2018), Tanzania (2014), and Uganda (2011).

LSMS surveys collect data in the same way as DHS surveys, and we similarly buffer the provided jittered LSMS cluster points in the same way, using a 2km radius for urban clusters and a 10km radius for rural clusters (see Section D.1). This buffering creates a polygon, which we use in the analysis.

Unlike the DHS surveys, the survey structure and the phrasing of specific questions in LSMS surveys are not consistent across countries. Because of this, each of the LSMS variables we use in this analysis could be calculated only for a subset of the survey countries listed above. We took care to ensure that construct validity for each variable is as consistent as possible across countries. For example, all variables given in currency units were converted from local currency to 2010 international dollars using a purchasing power parity exchange rate from the World Bank. Still there is some variation from country to country regarding how each variable is measured, with examples given below. For this reason, all LSMS variables are modeled using distinct country-level prediction models. Estimates from these various models are concatenated when their performance is presented in the text and figures. More information on the specific LSMS variables is given below:

- *Ag involvement.* This variable indicates that a household had at least one member involved in agriculture in the previous 12 months. There is considerable variation in how this question is asked (e.g., asking about cultivating any plot or asking separate questions regarding whether a household member has worked on either land that belongs to them or land that does not belong to them). We average household level binary responses to the LSMS cluster level and multiply by 100 to get a percentage. We estimate this variable for the following LSMS surveys: Ethiopia (2018), Niger (2011), Nigeria (2018), Tanzania (2014).
- *Cellphone ownership LSMS.* This variable indicates whether households have at least one mobile phone. A similar question is also asked in the DHS surveys. There is considerable variation in how this question is asked (e.g., some surveys are explicit that the mobile phone needs to be in working order). We average household level binary responses to the LSMS cluster level and multiply by 100 to get a percentage.

We estimate this variable for the following LSMS surveys: Ethiopia (2018), Malawi (2016), Niger (2011), Nigeria (2018), Tanzania (2014), and Uganda (2011).

- *Consumption.* This variable refers to the total value of household consumption per adult in each household. LSMS surveys give this aggregated value based on a number of more specific questions about household purchases. There is considerable variation in how this question is asked and the formulas for calculating household consumption are not always consistent across LSMS surveys. We average the values from surveyed households within each LSMS cluster. We calculate this metric for the following LSMS surveys: Ethiopia (2018), Malawi (2016), Niger (2011), Nigeria (2016), and Tanzania (2014).
- *Food worrying.* Many LSMS surveys ask whether a household has worried whether they would have enough food in the past 7 days. We calculate the percent of households in a LSMS cluster that responds in the affirmative to this binary question for the following set of LSMS countries: Ethiopia (2018), Malawi (2016), and Tanzania (2014). See ref. (87) for more information on food security and LSMS surveys.
- *Food coping.* LSMS food security questions use a 0 to 7 scale, asking about how many days in the past week a food security coping mechanism was employed (e.g., limiting the variety of food eaten). We use this to create a composite index that represents the extent to which food security coping is employed by each household. If a household employed all mechanisms asked about every day of the week, the value is a 7. If the household employed no food security coping mechanisms, the value is a 0. At the enumeration area level, this can be interpreted as the average level of food security coping within the enumeration area. These questions were available for the following LSMS surveys: Ethiopia (2018), Malawi (2016), Tanzania (2014), and Nigeria (2015). In Nigeria, households were asked these questions for the same year during planting and harvesting season and we average these responses. A closely related version of this food security index is described and validated by Bertelli (2019) (87).
- *Livestock ownership.* This variable indicates whether households had at least one member involved in raising livestock in the last 12 months. There is considerable variation in how this question is asked (e.g., directly or by asking specifically about various livestock types) indicating that this variable does not have consistency across country-level surveys. We average these binary responses to the LSMS cluster level. Livestock related questions are available for the following LSMS surveys: Ethiopia (2018), Malawi (2016), and Nigeria (2018).

- *Livestock sales.* This variable refers to the total value of all live or slaughtered livestock sales in the past 12 months. There is considerable variation in how this question is asked across countries (e.g., either directly or broken down by slaughtered and live animals). We average the values from surveyed households within each LSMS cluster. We calculate this metric for the following LSMS surveys: Malawi (2016) and Tanzania (2014).
- *Livestock value.* This variable refers to the total value in of all livestock sales owned by the household. There is considerable variation in how this data is collected across countries (e.g., asking about aggregate value or asking a number of questions broken down by livestock type). We average the values from surveyed households within each LSMS cluster. We calculate this metric for the following LSMS surveys: Ethiopia (2018), Malawi (2016), and Nigeria (2018).
- *Produce sales.* This variable stems from asking households about the total value of all agricultural sales in the last 12 months. There is considerable variation in how this data is collected across countries (e.g., either through a single question or via a variety of questions, broken down by crop type or season). We average the values from surveyed households within each LSMS cluster. Questions about produce sales were available in the following LSMS surveys: Ethiopia (2018), Malawi (2016), Niger (2011), Nigeria (2018), Tanzania (2014), and Uganda (2011).
- *Receiving loan.* This metric represents the percent of households within an enumeration area that received a loan in the 12 months prior to the survey question being asked. There is significant variation in how this question is asked across countries (e.g., sometimes a minimum threshold value is mentioned). We average these binary responses to the LSMS cluster level to create a percent value. Information about households obtaining loans could be constructed from the following LSMS surveys: Ethiopia (2018), Malawi (2016), Nigeria (2018), and Tanzania (2014).
- *Rejected for loan.* This variable represents the percent of households within an enumeration area that both applied for a loan and were subsequently rejected for a loan. There is significant variation in how this information is collected across countries (e.g., as a single, two-part question or broken into separate questions). Information on households being rejected for a loan were constructed from the following LSMS surveys: Ethiopia (2018), Malawi (2016), and Nigeria (2018).

Category	Source	Coverage	Spatial Structure	Spatial Resolution	Collection Years ¹	Variables
Agricultural Assets	(30)	DHS countries ²	Polygon	5km (urban) or 10km (rural) radii containing village	2012 - 2019	Ag land ownership; Livestock ownership
	(29)	LSMS countries ³	Polygon	5km (urban) or 10km (rural) radii containing village	2011 - 2019	Produce sales; Livestock value; Livestock sales
	(33)	Global	Raster	5 arc-minutes	2005	Global cropped area; Global maize cropped area; Global pasture area; Global rice cropped area; Global soy cropped area; Global wheat cropped area
	(34)	Global	Raster	5 arc-minutes	2005 - 2013	Global irrigated area
	(29)	LSMS countries ³	Polygon	5km (urban) or 10km (rural) radii containing village	2014 - 2019	Ag involvement
	(35)	USA, EU, BRA	Polygon	Admin 2 Polygons ⁴	1983 - 2009	Soy yield
	(36)	USA, EU, CHN, BRA	Polygon	Admin 2 Polygons ⁴	1983 - 2009	Maize yield
Built Infrastructure	(32)	USA	Raster	30m x 30m	2019	US maize cropped area; US cotton cropped area; US nontree cropped area; US soy cropped area; US tree cropped area; US wheat cropped area
			Polygon	Census tract	2019	Housing density; Building age; Housing units occupied; Housing units vacant

Category	Source	Coverage	Spatial Structure	Spatial Resolution	Collection Years ¹	Variables
	(38)	USA	Line	Pipeline path	2020	Crude oil pipelines; HGL pipelines; Petroleum pipelines
	(37)	Global	Polygon	Building footprint	2012	Building footprint area
	(39)	Global	Line	Road path	2020	Road length
	Demographics (32)	USA	Polygon	Census tract	2019	Food stamps; Households; Median age; Recent birth; Population
Education	(32)	USA	Polygon	Census tract	2019	Bachelors degree; Graduate degree; Less than high school degree
	(30)	DHS countries ²	Polygon	5km (urban) or 10km (rural) radii containing village	2012 - 2019	Higher edu; Primary edu; Secondary edu; Male literacy; Female literacy
	(31)	Global	Polygon	Admin 1 Polygons ⁵	2018	Expected schooling; Schooling
Health	(30)	DHS countries ²	Polygon	5km (urban) or 10km (rural) radii containing village	2012 - 2019	Child mortality; Child height-for-age; Infant mortality; Child malaria; Neonatal mortality; Child overweight; Child overweight-for-age; Child stunting; Child underweight-for-age; Child wasting; Child weight-for-age; Child weight-for-height
	(29)	LSMS countries ³	Polygon	5km (urban) or 10km (rural) radii containing village	2014 - 2019	Food coping; Food worrying
	(31)	Global	Polygon	Admin 1 Polygons ⁵	2018	Life expectancy
Household Assets	(32)	USA	Polygon	Census tract	2019	Bedrooms; Home value; Homes w LP gas; Computer ownership; Vehicle ownership

Category	Source	Coverage	Spatial Structure	Spatial Resolution	Collection Years ¹	Variables
	(30)	DHS countries ²	Polygon	5km (urban) or 10km (rural) radii containing village	2012 - 2020	Electricity access; Bike ownership; Car ownership; Improved water source; Floor material; Intl Wealth Index; Landline telephone; Cellphone ownership DHS; Refrigerator ownership; Flush toilet; Latrine; No sanitation facility; Sleeping rooms; TV ownership; Drinking water access
	(29)	LSMS countries ³	Polygon	5km (urban) or 10km (rural) radii containing village	2011 - 2019	Cellphone ownership LSMS
Income	(32)	USA	Polygon	Census tract	2019	Gross rent; Income; Income:poverty ratio
	(40)	Global	Raster	15 arc-seconds	2015	Nighttime lights radiance
	(29)	LSMS countries ³	Polygon	5km (urban) or 10km (rural) radii containing village	2014 - 2019	Consumption; Receiving loan; Rejected for loan
	(31)	Global	Polygon	Admin 1 Polygons ⁵	2018	Gross Natl Income; Human Dev Index
Natural Systems	(46)	Global	Raster	1km x 1km	2018	Ruggedness
	(2)	Global	Raster	30m x 30m	2010	Forest Cover
	(47)	Global	Polygon	.01° x .01°	1960 - 2018	Tree diversity
	(43)	Global	Raster	611.5m x 611.5m at the equator	2020	Elevation
	(44)	Global	Raster	0.1° x 0.1°	2003 - 2016	Gravitational anomaly
	(45)	Global	Raster	1-arc-degree equal area cells	2003 - 2016	Gravitational anomaly change

Category	Source	Coverage	Spatial Structure	Spatial Resolution	Collection Years ¹	Variables
	(42)	CA, USA	Polygon	39.2 km buffer at start point of survey	2019	Bird diversity
	(41)	USA	Point	Point	2009 - 2019	Bee diversity
Occupation	(32)	USA	Polygon	Census tract	2019	Empl in sales/office; Empl in service; Unemployment
	(30)	DHS countries ²	Polygon	5km (urban) or 10km (rural) radii containing village	2012 - 2019	Men empl in ag; Men empl in clerical; Men empl in domestic; Men empl in professional; Men empl in skilled labor; Men empl in unskilled labor; Women empl in ag; Women empl in clerical; Women empl in domestic; Women empl in professional; Women empl in service; Women empl in skilled labor; Women empl in unskilled labor

Table SM4: Data sources used to train 115 imagery-based models. Table lists all data sources used to train imagery-based models, grouped by 10 variable categories (agricultural assets, agriculture, built infrastructure, built infrastructure, demographics, education, health, household assets, income, natural systems, and occupation). Data characteristics reported in this table (spatial extent, spatial structure, resolution, and collection years) describe the data in their original form, before any processing is conducted for harmonization and model training.

¹ Table lists years of data available from each source. Throughout the analysis, data from the single year closest to 2019 are used to train imagery-based prediction models.

² Demographic and Health Surveys (DHS) countries include data from various countries in Sub-Saharan Africa, North Africa, West Asia, Europe, Central Asia, South Asia, Southeast Asia, Oceania, Latin America, and the Caribbean. Survey data availability varies across countries and variables.

³ Living Standards Measurement Survey (LSMS) countries include data from Ethiopia (2018), Malawi (2016), Niger (2011), Nigeria (2015 & 2018), Tanzania (2014), and Uganda (2011). Survey data availability varies across countries and variables.

⁴ “Admin 2 Polygons” refer to the second administrative unit in each country (e.g., counties in the United States or districts in India).

⁵ “Admin 1 Polygons” refer to the first administrative unit in each country (e.g., states in the United States and in India).

E Supplementary Text: Automated model specification

Below we describe the details of our automated model tuning approach. Throughout this process, all tuning decisions are made via 5-fold cross-validation by selecting the model with the best performance in the 80% validation set. Model performance is later evaluated in a 20% held-out test set. In all models, we restrict predictions to be within the range of observed ground truth values for each variable. Model tuning includes selection of: the value of the ridge regression penalization hyperparameter; whether to model the outcome in natural logarithms (logs) or levels; whether to use a globally homogeneous model or continent-specific model; whether to include an intercept in the model; and whether to weight features by area or by population when spatially aggregating.

To reduce computation, we split model tuning into two stages. We first tune whether the variable should be modeled in logs or in levels and we then tune the remaining model hyperparameters. We tune the ridge hyperparameter in all models, conducting the search by testing values in the direction of progressively increasing R^2 until an optimum is found, or until the marginal improvement to the R^2 in the validation set is less than 0.001. We detail this two-stage procedure in the following subsections.

E.1 First stage

Log or level In the first stage of tuning we decide whether the variable is best modeled directly in levels or after taking the natural logarithm of the variable value. Taking the natural logarithm of a variable can help if satellite imagery explains percentage changes in the variable better than they do level changes and/or if the variable’s distribution in the ground truth data is highly skewed. Because levels are usually easier to interpret, we implement this stage using a bias toward selection of the levels model: specifically, we only decide to model the variable in logs if the R^2 is least 10% higher in this model than it is in levels. In this stage we implement both the models in logs and levels with the following specifications for other model components: continent-specific models (if the variable has ground truth data spanning multiple continents); including an intercept; and weighting the features by area when spatially aggregating. We then tune these other model parameters in the second stage. This selection procedure resulted in approximately one third of the variables being modeled in logs.

When modelling variables in logs that have some values that are negative or zero, we add a small constant to all values. For variables that contain negative values, we add the absolute

value of the minimum to make all values positive. And then for variables that contain zeros, we add the first percentile of the non-zero data.

E.2 Second stage

With the decision of whether to model each variable in logs or levels held fixed, we then run all possible combinations of the remaining tuning options and select the one with the best out-of-sample performance within the 80% validation set. Each model tuning decision is described in detail below.

Globally homogeneous or continent-specific model For variables that span multiple continents and have at least 20 observations in each continent, we estimate both a globally-homogeneous and a continent-specific model. Continent assignments come from the ESRI World Continents shapefile.⁵ We train a separate model for each continent, allowing each to have its own optimal ridge hyperparameter. For the variables in which both global and continent-specific models were run, our autotuning selection chooses the continent-based model 98% of the time. Only one variable (women employed in clerical) was best modeled in terms of a globally homogeneous model when a continent model was tried. For variables estimated with continent-specific models, overall model performance is reported by pooling predictions and ground truth observations across all continents and estimating the coefficient of determination (i.e., R^2) in the pooled data.

For variables derived from the Living Standards Measurement Survey (LSMS) (29) (see Table D.20), we exclusively train country-specific models – not globally homogeneous or continent-specific models – due to differences across countries in the way the survey data are structured (Supplementary Materials Section D.20). Model performance pooled across all countries is reported analogously to pooled performance across continents.

Population-weighted or area-weighted feature aggregation For variables in which MOSAIKS imagery features are spatially aggregated to match the resolution of the ground truth measurement (i.e., polygon variables), we estimate models using both area-weighted feature averages and population-weighted feature averages. We use population density weights from the Gridded Population of the World V4 (GPW) (51) to compute population-weighted average values. Missing population density observations from GPW are assumed to have the minimum population density weight (1×10^{-6}). In the set of variables for which both options were run, models averaging features weighting by population achieved better

⁵Available at: <https://hub.arcgis.com/datasets/esri::world-continents/about/>.

performance than models weighting by area 54% of the time. Intuitively, no variables in the agriculture or natural systems categories benefited from population weighting.

Intercept or no intercept Consistent with previous MOSAIKS research (7), we observe that the inclusion of an intercept term is not strictly necessary. We estimate each model specification with and without an intercept term and find that including an intercept term improves performance 85% of the time. In continent-based models and in LSMS variables, each continent-specific or country-specific model has its own intercept.

F Additional Supplementary Figures: Spatial extrapolation

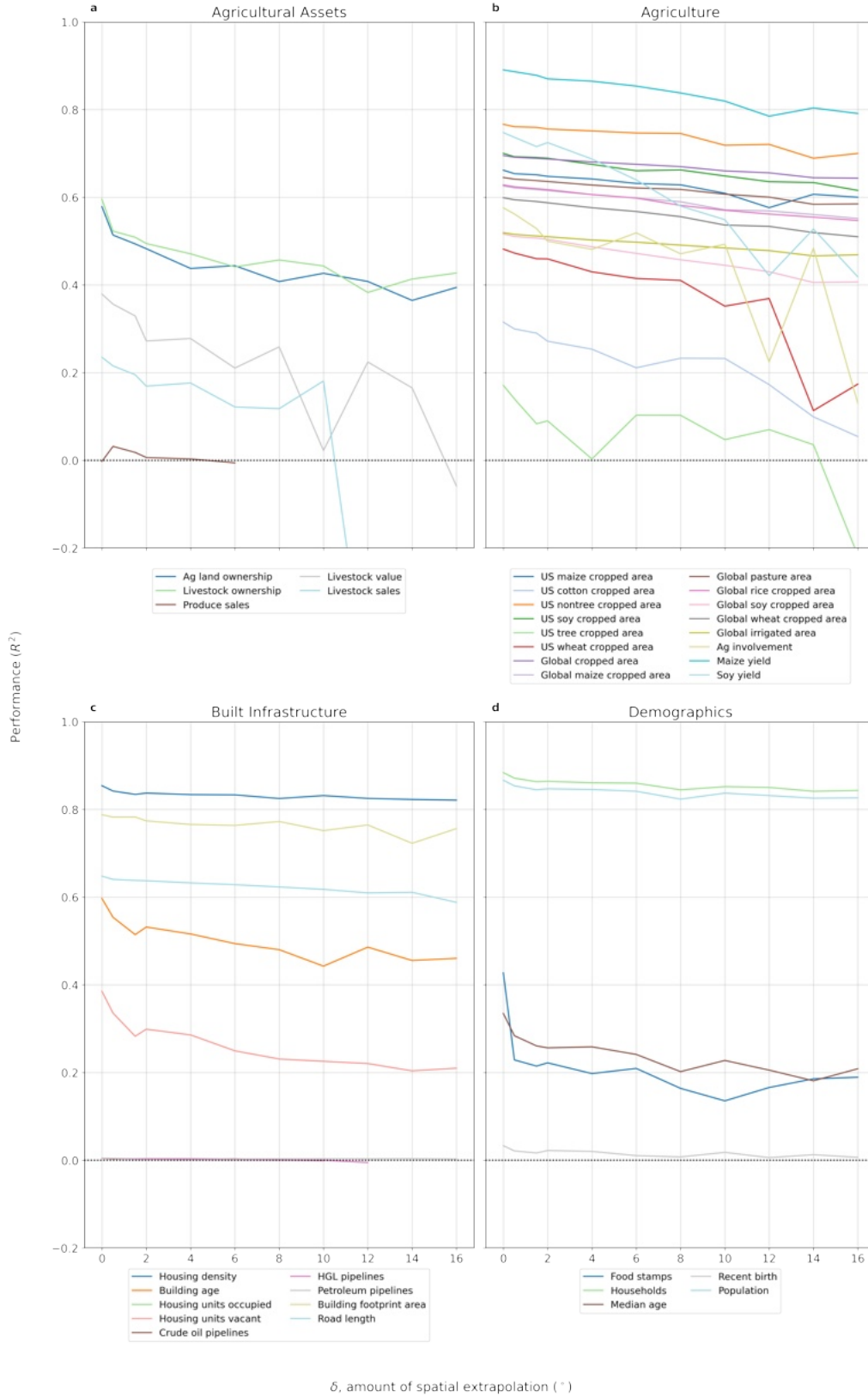


Figure SM9: Model performance far from training data: agricultural assets, agriculture, built infrastructure, and demographics. Identical to Figure 3, with added labels indicating which of the 115 variables studied corresponds to each line shown.

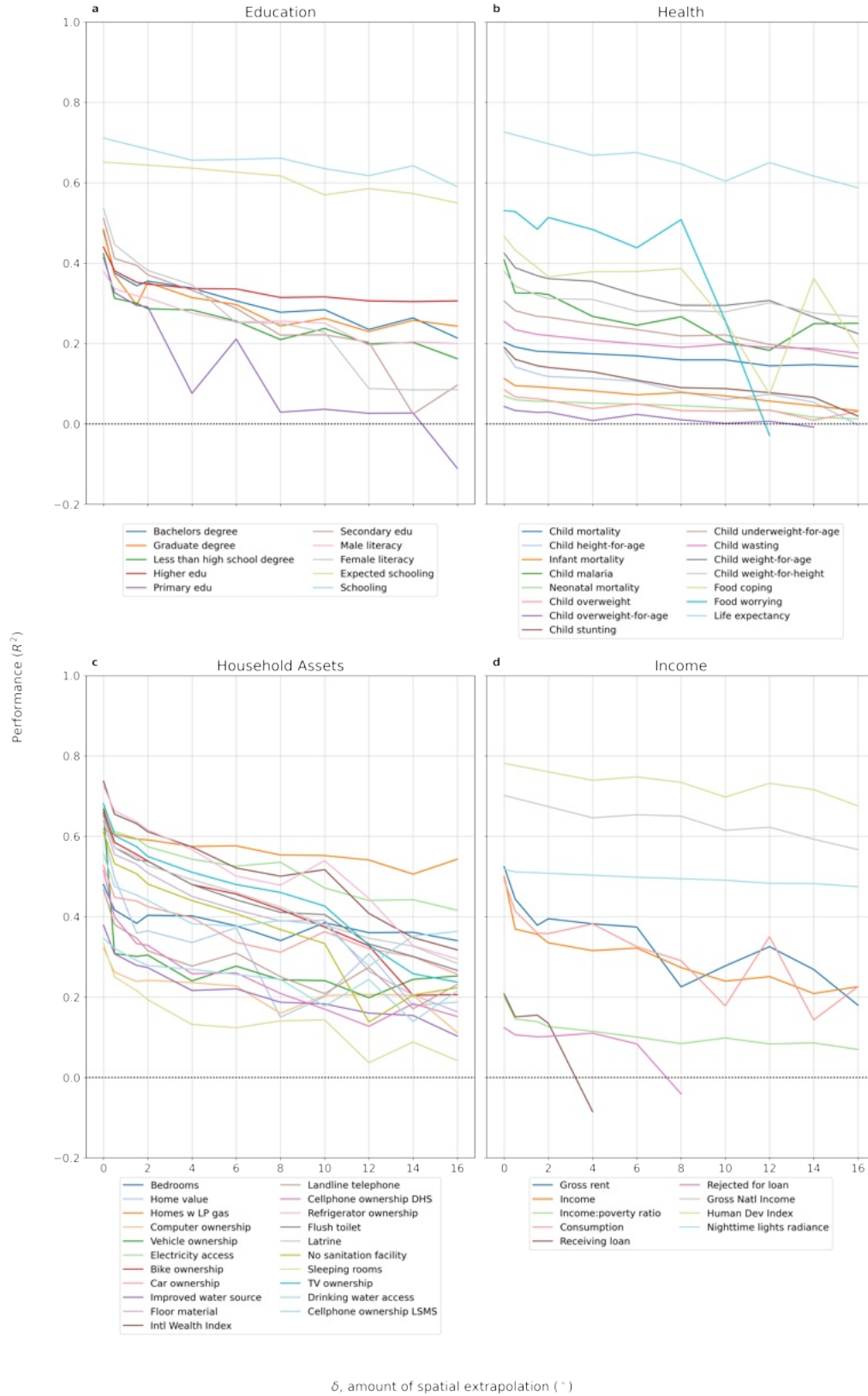
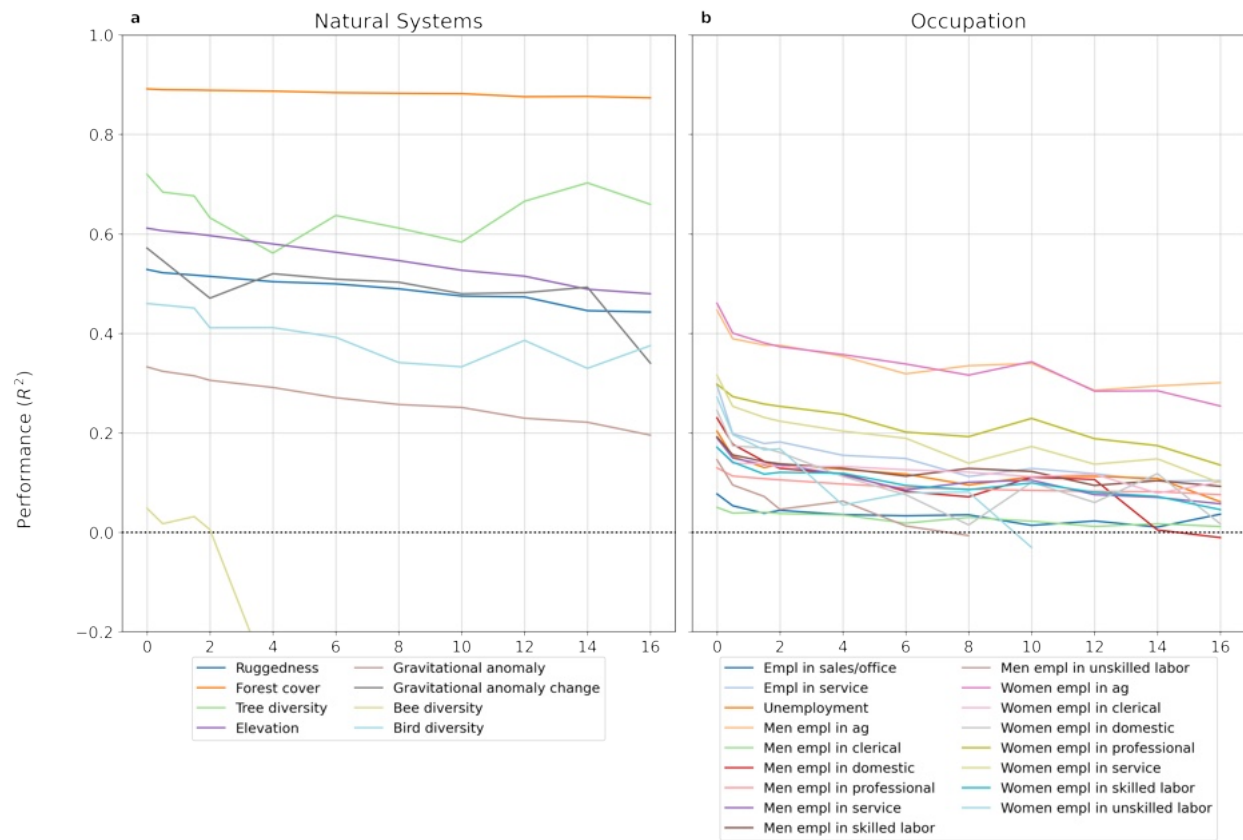


Figure SM10: Model performance far from training data: education, health, household assets, and income. Identical to Figure 3, with added labels indicating which of the 115 variables studied corresponds to each line shown.



δ, amount of spatial extrapolation (°)

Figure SM11: Model performance far from training data: natural systems and occupation. Identical to Figure 3, with added labels indicating which of the 115 variables studied corresponds to each line shown.

G Additional Supplementary Figures: Error analysis by category

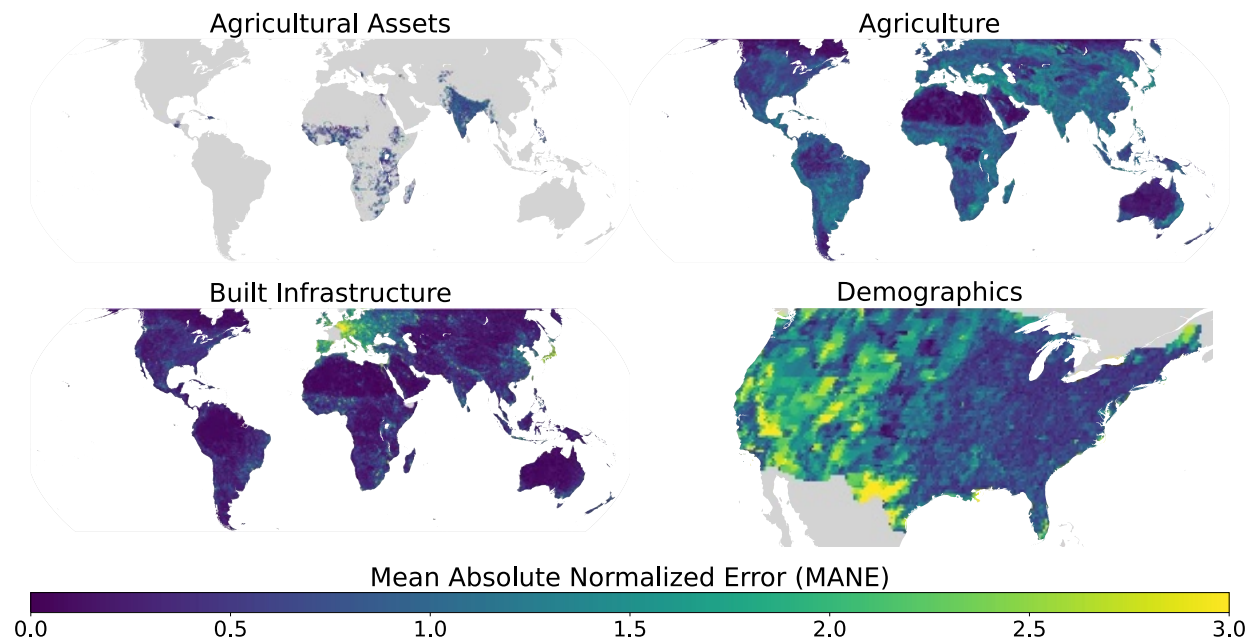


Figure SM12: MANE for agricultural assets, agriculture, built infrastructure, and demographics. Identical to figure 4, except MANE is calculated and plotted by variable category. We predict human labels only in populated locations. Locations where there is no value of MANE for that category are plotted in light gray.

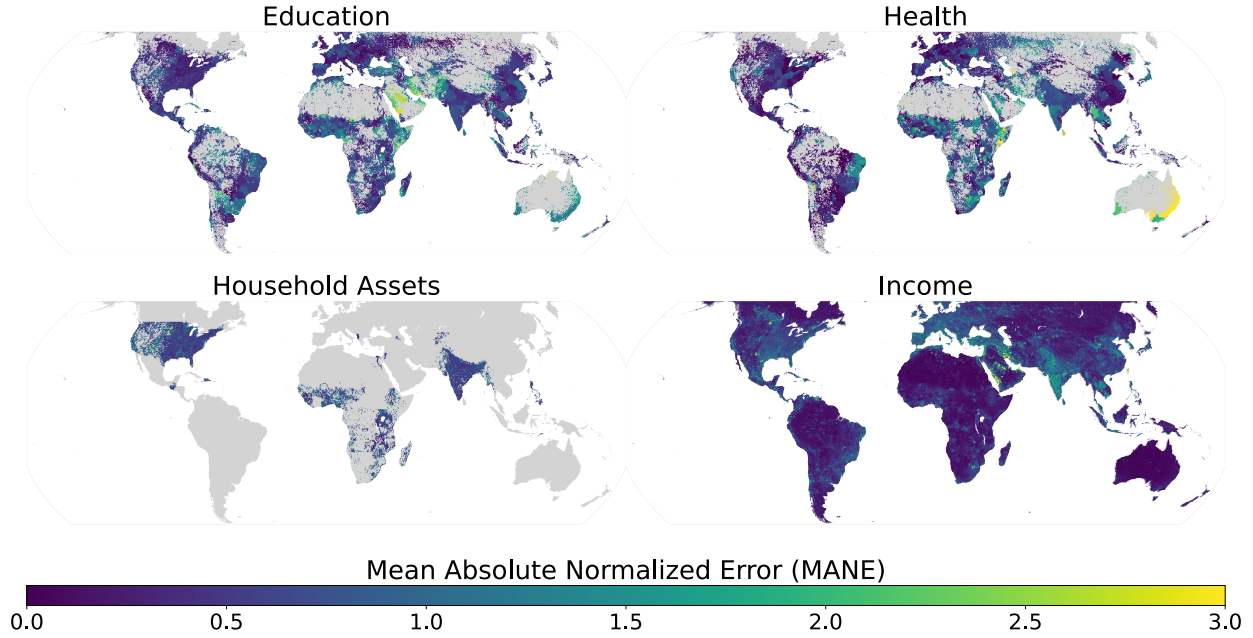


Figure SM13: MANE for education, health, household assets, and income. Identical to figure 4, except MANE is calculated and plotted by variable category. We predict human labels only in populated locations. Locations where there is no value of MANE for that category are plotted in light gray.

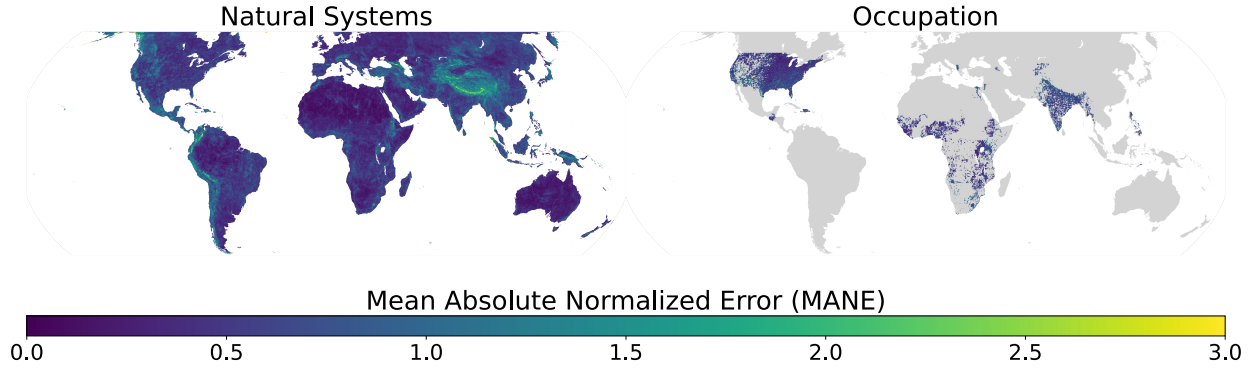


Figure SM14: MANE for natural systems and occupation. Identical to figure 4, except MANE is calculated and plotted by variable category. We predict human labels only in populated locations. Locations where there is no value of MANE for that category are plotted in light gray.